Elements of Statistical Inference

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Statistical inference mainly deals with estimation and hypothesis testing about unknown population parameters, given a set of observations on the variable whose population behaviour we want to study or model. Using the developed model for forecasting or prediction purpose also falls under the realm of statistical inference but its general theory will not be discussed here and will be introduced only when it first arises in the context of regression. Here we shall only give the flavour of general mathematical statistical treatment of the problem of estimation and hypothesis testing about unknown population parameters.

1 Some Terminologies

1.1 Parameter

Parameters are quantities defined in the population. To emphasise this fact we shall refer to it as population parameter in this subsection. Population parameters are quantities derived from the population probability model like its mean, median, variance, 90-th percentile, or even the entire c.d.f. itself. In practice we often assume observations arising from a given parametric family of probability model like Binomial, Geometric, Negative Binomial, Poisson, exponential, Gamma, Normal, Weibull etc. from external or empirical considerations. In such situations population parameters are nothing but those quantities which appear in the expressions of the p.m.f. or p.d.f. of those probability models, like the p of Binomial, Geometric or Negative Binomial distribution, or λ of the Poisson or exponential model or (μ, σ^2) of the Normal probability model, or (α, λ) of the Gamma distribution or (λ, β) of the Weibull model. This is because in these parametric families any other population quantity of interest can be expressed in terms of these basic model parameters. For instance the standard deviation of a Bernoulli population is $\sqrt{p(1-p)}$, the mean of a Gamma population is α/λ, or the 75-th percentile of a Normal population is $\mu + 0.6745 \times \sigma$.

1.2 Statistic

Statistics are quantities which are computed from the data or observations. For example for a random sample $Y_1, Y_2, \ldots, Y_n$ from a $N(\mu, \sigma^2)$ population, the sample mean $\overline{Y} = \frac{1}{n}\sum_{i=1}^n Y_i$ or the naive sample variance $s_n^2 = \frac{1}{n}\sum_{i=1}^n (Y_i - \overline{Y})^2$ are statistics, as opposed to their respective population counter-parts μ and σ^2, which are population parameters. Sample proportion $\hat{p}$ = (# Successes in the sample)/n is another example of a statistic with a sample of size n from a Bernoulli 0-1 or Success-Failure population. In general a statistic is a function of observations $Y_1, Y_2, \ldots, Y_n$ which is denoted by $T(Y_1, Y_2, \ldots, Y_n)$, to emphasise the fact that it is a formula used to derive quantities based on a sample $Y_1, Y_2, \ldots, Y_n$.

1.3 Sampling Distribution

In classical statistics the optimality of any method is judged in terms of its repeated use over different samples from the same population. This philosophy gives rise to the consideration of the possible values a statistic $T(Y_1, Y_2, \ldots, Y_n)$ can take and the frequency with which it takes such values over repeated sampling. Or in other words we consider the probability distribution of a statistic $T(Y_1, Y_2, \ldots, Y_n)$ over repeated sampling. This probability distribution is called the sampling distribution of T.

For example consider the sample mean $\overline{Y}$. Though for a given population its mean μ is something fixed (but unknown), we cannot expect to get the same value of the sample mean $\overline{Y}$ for all different possible samples that we can draw from the population. However we can consider and theoretically derive how $\overline{Y}$ would behave over repeated sampling for all possible samples in terms of its probability distribution. This probability distribution of $\overline{Y}$ over all possible samples is called the sampling distribution of the sample mean $\overline{Y}$.

Example 1: Consider a population consisting of only four numbers 1, 2, 3 and 4 with 30% 1's, 40% 2's, 20% 3's and the remaining 10% 4's. That is the probability distribution in the population may be expressed in terms of the p.m.f. $p_Y(y)$ of the population random variable Y as follows:

y	1	2	3	4
$p_Y(y)$	0.3	0.4	0.2	0.1

Now consider drawing a random sample of size 2 from this population and the three statistics $\overline{Y} = (Y_1 + Y_2)/2$, $s_2^2 = 1/2\sum_{i=1}^{2}(Y_i - \overline{Y})^2$ and $s_1^2 = \sum_{i=1}^{2}(Y_i - \overline{Y})^2$, where Y_1 and Y_2 are the two observations. The sampling distributions of these three statistics can be figured out by considering all possible samples of size 2 that can be drawn from this population, the corresponding probabilities of drawing each such sample, and the values of each of the statistics for every such sample These steps are presented in the following table:

Possible Samples	{1,1}	{1,2}	{1,3}	{1,4}	{2,2}	{2,3}	{2,4}	{3,3}	{3,4}	{4,4}
Probability	0.09	0.24	0.12	0.06	0.16	0.16	0.08	0.04	0.04	0.01
$\overline{Y}$	1	1.5	2	2.5	2	2.5	3	3	3.5	4
s_2^2	0	0.25	1	2.25	0	0.25	1	0	0.25	0
s_1^2	0	0.5	2	4.5	0	0.5	2	0	0.5	0

Finally consolidating the distinct values that have been assumed by these statistics and adding the corresponding probabilities of obtaining the samples for the same values of the statistics, we obtain the sampling distributions of these statistics as follows:

Sampling Distribution of the Sample Mean $\overline{Y}$

$\overline{y}$	1.0	1.5	2.0	2.5	3.0	3.5	4.0
$p_{\overline{Y}}(\overline{y})$	0.09	0.24	0.28	0.22	0.12	0.04	0.01

Sampling Distribution of s_2^2

s_2^2	0	0.25	1	2.25
prob.	0.30	0.44	0.20	0.06

Sampling Distribution of s_1^2

s_2^2	0	0.5	2	4.5
prob.	0.30	0.44	0.20	0.06

2 Estimation

Suppose we have a probability model in the population characterised by the p.d.f. $f(y|\theta)$ (or p.m.f. $p(y|\theta)$) and we have n independent and identically distributed (henceforth called i.i.d.) observations $Y_1, Y_2, \ldots, Y_n$ on the population random variable Y, which has the p.d.f. $f(y|\theta)$ (or p.m.f. $p(y|\theta)$). The first inference problem at hand is to estimate the unknown population parameter θ. The problem of estimation has two fangs - point estimation and interval estimation. In the first case *i.e.* point estimation, as the name suggests, one deals with a single valued estimator of θ; while in the later case *i.e.* interval estimation, one reports an interval of values which is supposed to contain the true unknown value of the parameter θ.

2.1 Point Estimation

We start our discussion on point estimation by first considering the nature of a "good" estimator. That is we shall first try to understand the kind of properties and behaviour a reasonable estimator should have. Or in other words we first discuss the desirable criteria for "good" estimators. Then failing to devise any straight-forward method of obtaining such estimators we shall next resort to some general methods of estimation which are reasonably well-behaved for large samples for the so called "regular" models.

Let the statistics $\hat{\theta}$ and $\hat{\theta}'$ be two different estimators of the population parameter θ. Say for example with a sample from a $N(\mu, 1)$ population, for the unknown population parameter μ, let $\hat{\mu}$ be the sample mean and $\hat{\mu}'$ be the sample median, as μ is both the population mean as well as the population median. Likewise with a sample from a Poisson(λ) population, for the unknown population parameter λ, let $\hat{\lambda}$ denote the sample mean and $\hat{\lambda}'$ denote the sample variance, as λ may be interpreted as both the population mean and population variance. Now $\hat{\theta}$ would be considered to be better than $\hat{\theta}'$ if

$$\text{Prob}_\theta(\theta - a < \hat{\theta} < \theta + b) \geq \text{Prob}_\theta(\theta - a < \hat{\theta}' < \theta + b) \quad \forall a, b > 0 \text{ and } \forall \theta \in \Theta \tag{1}$$

where the probabilities above are according to the sampling distributions of $\hat{\theta}$ and $\hat{\theta}'$ (it is subscripted with a θ to emphasise the fact that these probabilities in general depend on the unknown θ), and Θ is the **parameter space**, the set of all possible values the unknown population parameter θ can take. A slightly weaker version of inequality (1) is

$$E_\theta[(\hat{\theta} - \theta)^2] \leq E_\theta[(\hat{\theta}' - \theta)^2] \quad \forall \theta \in \Theta \tag{2}$$

where the expectation $E_\theta[\cdot]$ is w.r.t.the sampling distributions of the estimators $\hat{\theta}$ and $\hat{\theta}'$ (again the $E_\theta[\cdot]$ is subscripted with a θ to emphasise the fact that the expectation in general

depends on the unknown θ). Inequality (2) is weaker than inequality (1) in the sense that if inequality (1) is satisfied by two estimators $\hat{\theta}$ and $\hat{\theta}'$, then it implies that they must satisfy inequality (2). Since inequality (2), though weaker, is less clumsy to deal with, as it does not involve additional arbitrary constants like a and b as in inequality (1), "goodness" of an estimator is measured in terms of this weaker criterion. This criterion is called the Mean Squared Error or MSE which is formally defined as follows.

<u>Definition 1:</u> Mean Squared Error or **MSE** of an estimator $\hat{\theta}$ is given by $\text{MSE}_\theta(\hat{\theta}) = E_\theta[(\hat{\theta} - \theta)^2]$.

Intuitively, if $\hat{\theta}$ is an estimator of θ, then the *error* of $\hat{\theta}$ is $(\hat{\theta} - \theta)$, the amount by which it misses the quantity θ it is trying to estimate. But this error could be either positive or negative and in general we are not interested in its sign but only the magnitude. A smooth way of getting rid of the sign of a quantity is to square it.[1] Thus we consider the *squared error* $(\hat{\theta} - \theta)^2$. Now this *squared error* is a random quantity as its value depends on the value of the estimator $\hat{\theta}$, which varies from sample to sample. One way of consolidating the value of this random criterion *viz. squared error* would be to look at its mean over repeated sampling or by how much is the estimator $\hat{\theta}$ missing its target value of θ on an average. This leads to the criterion *mean squared error* $E_\theta[(\hat{\theta} - \theta)^2]$.

Note that though MSE of an estimator $\hat{\theta}$ does not depend on the value of $\hat{\theta}$ (since we have averaged it out), in general it still depends on θ. Thus MSE of $\hat{\theta}$ is a function of θ. Since we do not know the value of θ, which is the crux of the problem, one may now seek for a super estimator which has the smallest MSE among all the estimators of θ, no matter what the value of θ may be or *for all values of* $\theta \in \Theta$. Unfortunately such super estimators do not exist, because when $\theta = \theta_0$, some fixed value in Θ, no other estimator $\hat{\theta}$ can have a smaller MSE than the (rather silly) estimator $\hat{\theta}' \equiv \theta_0$. That is if we allow anything and everything in the universe to be an estimator of θ and then seek for an estimator which uniformly minimises the MSE, that is simply asking for too much.

The solution becomes apparent once we have realised the problem. If we can somehow keep trivial and silly estimators out of the consideration and then seek for an MSE minimising estimator, there might be some hope of obtaining a solution. That is we have to keep our search for a "good" or MSE minimising estimator confined within a narrower class of "reasonable" estimators. One such smaller or restricted class of estimators which is usually considered in practice are called **unbiased** estimators, defined as follows.

<u>Definition 2:</u> An estimator $\hat{\theta}$ of the unknown population parameter θ is called **unbiased**, if $E_\theta[\hat{\theta}] = \theta \ \forall \theta \in \Theta$.

[1] Getting rid of the sign of a quantity is a recurring problem in statistics and we always use squaring for accomplishing this task. Simply ignoring the sign is not a smooth procedure as the graph of $f(x) = |x|$ for $-\infty < x < \infty$ exhibits at $x = 0$. The closest one can come to this through a smooth procedure is by squaring it (study the graph of $g(x) = x^2$ for $-\infty < x < \infty$). Raising the quantity, whose sign we want to get rid of, by any even power would also be a smooth procedure but the magnitude is distorted the least when this power is 2 (for instance compare the graphs of $f(x) = |x|$, $g(x) = x^2$ and $h(x) = x^4$ for $-\infty < x < \infty$). We want the process of ridding the sign to be smooth because otherwise it poses mathematical difficulties like not being able to use standard calculas methods for subsequent theoretical development.

Above definition requires the mean of the sampling distribution of an unbiased estimator to coincide with the unknown population parameter it is trying to estimate, for its all possible values. Intuitively, this means that if an estimator is unbiased, on an average it will always hit the target, no matter what or where the target is. This looks like a fairly reasonable thing to demand from an estimator. Also notice that the unbiasedness requirement also gets rid of trivial constant estimators, as clearly an estimator like $\hat{\theta} \equiv \theta_0$ has an expected value of θ_0, which does not coincide with the value of the unknown population parameter θ for any value other than θ_0 in Θ.

Example 2: Suppose $Y_1, Y_2, \ldots, Y_n$ is a random sample from an arbitrary population with mean μ. Then since $E[Y_i] = \mu \ \forall \mu$ and $\forall i$, Y_5 is an unbiased estimator of μ, so is $(Y_2 + Y_3)/2$, so is $0.2Y_8 + 0.5Y_{11} + 0.3Y_{13}$ and so is of course the sample mean $\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$. $\bigtriangledown$

The above example goes to show that usually the class of unbiased estimators is quite large. Now the task is to seek for an estimator with minimum MSE among this reasonably large class of unbiased estimators. Unbiased estimators has a simple neatly interpretable expression for their MSE as the following corollary shows.

Theorem 1: $\text{MSE}_\theta(\hat{\theta}\ ) = V_\theta[\hat{\theta}\ ] + B_\theta(\hat{\theta}\ )^2$, where $V_\theta[\hat{\theta}\ ]$ is the variance of the estimator $\hat{\theta}$ w.r.t. its sampling distribution and $B_\theta(\hat{\theta}\ ) = E_\theta[\hat{\theta}\ - \theta]$ is the bias of the estimator $\hat{\theta}$.

Proof:

$$
\begin{aligned}
& MSE_\theta(\hat{\theta}\ ) \\
&= E_\theta[(\hat{\theta}\ - \theta)^2] \\
&= E_\theta[\{(\hat{\theta}\ - E_\theta[\hat{\theta}\ ]) + (E_\theta[\hat{\theta}\ ] - \theta)\}^2] \\
&= E_\theta[(\hat{\theta}\ - E_\theta[\hat{\theta}\ ])^2] + (E_\theta[\hat{\theta}\ - \theta])^2 + 2E_\theta[(\hat{\theta}\ - E_\theta[\hat{\theta}\ ])(E_\theta[\hat{\theta}\ ] - \theta)] \\
&\quad \text{(because } E_\theta[\theta] = \theta \text{ and } (E_\theta[\hat{\theta}\ - \theta])^2 \text{ is a constant, whose expectation is the constant} \\
&\quad \text{itself.)} \\
&= V_\theta[\hat{\theta}\ ] + B_\theta(\hat{\theta}\ )^2 + 2(E_\theta[\hat{\theta}\ ] - \theta)E_\theta[\hat{\theta}\ - E_\theta[\hat{\theta}\ ]] \\
&\quad \text{(by the definition of } V_\theta[\hat{\theta}\ ] \text{ and } B_\theta(\hat{\theta}\ ) \text{ and constancy of } E_\theta[\hat{\theta}\ ] - \theta) \\
&= V_\theta[\hat{\theta}\ ] + B_\theta(\hat{\theta}\ )^2 \ (\text{as } E_\theta[\hat{\theta}\ - E_\theta[\hat{\theta}\ ]] = 0)
\end{aligned}
$$

$\bigtriangledown$

Corollary 1: If $\hat{\theta}$ is unbiased for θ, $\text{MSE}_\theta(\hat{\theta}\ ) = V_\theta[\hat{\theta}\ ]$.

Proof: If $\hat{\theta}$ is unbiased for θ, its bias $B_\theta(\hat{\theta}\ ) = E_\theta[\hat{\theta}\ - \theta] = 0$, by the definition of unbiasedness, and the result follows from Theorem 1. $\bigtriangledown$

For the above corollary it is usually customary to report the value of an (unbiased) point estimator together with its $\sqrt{V_\theta[\hat{\theta}\ ]}$, called its **standard error**. Corollary 1 reduces the task of seeking an MSE minimising estimator among the class of unbiased estimators to that of seeking an unbiased estimator with a uniformly minimum variance or standard error. This leads to the discussion of obtaining in general the "best" point estimators called the Uniformly Minimum Variance Unbiased Estimators or UMVUE.

2.1.1 UMVUE

The foregoing discussion goes to show the desirability of having an UMVUE. But unfortunately there is no direct method of obtaining an UMVUE for any parameter of a given population probability model. By a direct method we mean a ready made algorithmic technique which readily yields a (may be a computerised numerical) solution as soon as the problem of obtaining an UMVUE is stated or formulated. Instead there are a couple of theorems which help one show or prove an estimator to be UMVUE. However before discussing these theorems it is first necessary though to introduce a couple of more concepts and results, without which the main theorems regarding UMVUE would remain inaccessible. The first such concept is called **sufficiency**.

Definition 3: A statistic $T(Y_1, Y_2, \ldots, Y_n)$ is said to be **sufficient** for θ if the conditional distribution of the original observations $Y_1, Y_2, \ldots, Y_n$ given $T(Y_1, Y_2, \ldots, Y_n) = t$ does not depend on θ.

Sufficiency plays a central role in mathematical statistics, not just for UMVUE. Intuitively, sufficient statistics provide a way of reducing the data without loosing any information about the unknown parameter θ. This is because if one has the value of the sufficient statistic T but the original data set $Y_1, Y_2, \ldots, Y_n$ is lost, one can still reconstruct a set of $Y_1, Y_2, \ldots, Y_n$ (using for example a random number generator) as it does not require knowledge of the unknown θ (by definition), which is *equivalent* to the original data set in the sense that its probability distribution remains the same as the original data set. Thus if sufficient statistics exist, one need not carry around the entire original raw data set for drawing inference about the model parameters. Just having the values of the sufficient statistics is good enough or sufficient, as these statistics carry all the relevant information about θ contained in the observations $Y_1, Y_2, \ldots, Y_n$.

Example 3: Suppose Y_1 and Y_2 are i.i.d. Poisson(λ) *i.e.* we have a sample of size 2 from a Poisson population. Consider the statistic $T = Y_1 + Y_2$.

$$
\begin{aligned}
& P(Y_1 = y | T = t) \\
&= \frac{P(Y_1 = y, Y_2 = t - y)}{P(T = t)} \\
&= \frac{P(Y_1 = y)P(Y_2 = t - y)}{e^{-2\lambda}(2\lambda)^t/t!} \\
&\quad \text{(because } Y_1 \text{ and } Y_2 \text{ are independent and } T \sim \text{Poisson}(2\lambda)) \\
&= \frac{e^{-2\lambda}\lambda^{y+t-y}/(y!(t-y)!)}{e^{-2\lambda}(2\lambda)^t/t!} \\
&= \binom{t}{y}\left(\frac{1}{2}\right)^y\left(\frac{1}{2}\right)^{t-y},
\end{aligned}
$$

which does not depend on the unknown population parameter λ. It should now be easy to see that if we had $Y_1, Y_2, \ldots, Y_n$ a sample of size n from a Poisson(λ) population and

$T = Y_1 + Y_2 + \cdots Y_n$,

$$P(Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n | T = t) = \frac{t!}{y_1! y_2! \cdots y_n!} \left(\frac{1}{n}\right)^{y_1} \left(\frac{1}{n}\right)^{y_2} \cdots \left(\frac{1}{n}\right)^{y_n},$$

which does not depend on λ. Thus according to the definition $T = \sum_{i=1}^n Y_i$ is a sufficient statistic for a Poisson sample. This is because if one has the value of T as t, one can reconstruct a version of the original sample by generating a set of values from a Multinomial$(t; \frac{1}{n}, \ldots, \frac{1}{n})$ distribution, without bothering to carry around all the n $Y_1, Y_2, \ldots, Y_n$ values. $\bigtriangledown$

Now trying to intuitively guess, obtain and then show a statistic is sufficient from definition, as has been done in example 3 above, is an arduous if not an impossible task. Fortunately there is a theorem, called the **Factorisation Theorem**, which helps one obtain a sufficient statistics in a routine manner from the expression of the p.m.f./p.d.f. of a probability model. At this point it will also be wise to broaden our horizon by considering all the unknown parameters in the population at once. Thus from now on we shall use a bold-faced $\boldsymbol{\theta}$ to denote the vector (more than one) of unknown parameters, while preserving the notation θ in case of a single unknown.[2] Before presenting the Factorisation Theorem theorem we need to introduce another extremely important statistical concept called the **likelihood function**.

Definition 4: If $Y_1, Y_2, \ldots, Y_n$ are i.i.d. with p.d.f. $f(y|\boldsymbol{\theta})$ (or p.m.f. $p(y|\boldsymbol{\theta})$) with realised values $Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n$ the **likelihood function** of $\boldsymbol{\theta}$ is given by $L(\boldsymbol{\theta}|y_1, y_2, \ldots, y_n) = \prod_{i=1}^n f(y_i|\theta)$ (or $\prod_{i=1}^n p(y_i|\theta)$ in the discrete case).

Very loosely speaking the likelihood function sort of gives the probability of observing the data at hand given a value of the model parameter $\boldsymbol{\theta}$. But since $\boldsymbol{\theta}$ is unknown, we try to view this quantity in its totality as a function of the unknown $\boldsymbol{\theta}$ as it varies over its domain $\boldsymbol{\Theta}$. It is important to realise that in the expression of the likelihood function, the variable of interest is $\boldsymbol{\theta}$ and not the observed data $Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n$. It is something akin to a probability only when viewed as a function of $y_1, y_2, \ldots, y_n$, but since the likelihood must be viewed as a function of $\boldsymbol{\theta}$ it is not a probability.

Example 4: **A.** If $Y_1, Y_2, \ldots, Y_n$ are i.i.d. $N(\mu, \sigma^2)$, with realised values $Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n$, then the likelihood function of (μ, σ^2) is given by

$$L(\mu, \sigma^2|y_1, y_2, \ldots, y_n) = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^n (y_i-\mu)^2} = (2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\left\{\sum_{i=1}^n (y_i-\overline{y})^2 + n(\overline{y}-\mu)^2\right\}} \tag{3}$$

B. If $Y_1, Y_2, \ldots, Y_n$ are i.i.d. Bernoulli(p), so that each Y_i is 0-1 valued with probability of assuming the value 1 is p and 0 is $1-p$, which may be expressed as the p.m.f. $p(y|p) = p^y(1-p)^{1-y}$ for $y = 0, 1$, with realised values $Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n$, then the likelihood function of p is given by

$$L(p|y_1, y_2, \ldots, y_n) = p^{\sum_{i=1}^n y_i}(1-p)^{n-\sum_{i=1}^n y_i} \tag{4}$$

[2]This convention of using bold-face for a vector and ordinary font for a scalar - be it the parameters or statistics - will be used through out these notes.

C. If $Y_1, Y_2, \ldots, Y_n$ are i.i.d. Poisson(λ) with realised values $Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n$, then the likelihood function of λ is given by

$$L(\lambda|y_1, y_2, \ldots, y_n\ ) = e^{-n\lambda}\lambda^{\sum_{i=1}^n y_i} / \prod_{i=1}^n y_i! \tag{5}$$

<u>**Theorem 2**</u> **(Factorisation Theorem):** If the population random variable Y has p.d.f. $f(y|\boldsymbol{\theta})$ (or p.m.f. $p(y|\boldsymbol{\theta})$) then given the the observed data $Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n$, statistics $\boldsymbol{T}(Y_1, Y_2, \ldots, Y_n\ )$ is sufficient for $\boldsymbol{\theta}$ if and only if the likelihood function can be factored as ($\boldsymbol{t}$ being the realised value of $\boldsymbol{T}$)

$$L(\boldsymbol{\theta}|y_1, y_2, \ldots, y_n\ ) = g(\boldsymbol{t}(y_1, y_2, \ldots, y_n\ ), \boldsymbol{\theta})h(y_1, y_2, \ldots, y_n\ ). \tag{6}$$

That is $\boldsymbol{t}(y_1, y_2, \ldots, y_n\ )$ is sufficient for $\boldsymbol{\theta} \Longleftrightarrow$ the likelihood function can be factored into two components, where the expression of the first component involves $\boldsymbol{\theta}$ and terms involving $y_1, y_2, \ldots, y_n$ appearing only through $\boldsymbol{t}(y_1, y_2, \ldots, y_n\ )$, and the expression of the second component involves only $y_1, y_2, \ldots, y_n$ without any term involving $\boldsymbol{\theta}$.

<u>**Proof:**</u> We shall present the proof for discrete Y, which is a little more intuitive and illustrative but less technical than the continuous case.
<u>"only if" or $\Longrightarrow$ part:</u> Suppose $\boldsymbol{T}(Y_1, Y_2, \ldots, Y_n\ )$ is sufficient for $\boldsymbol{\theta}$ and let $\boldsymbol{t}(y_1, y_2, \ldots, y_n\ ) = \boldsymbol{t}$ denote the observed value of $\boldsymbol{T}$. Then

$$\begin{aligned}
&L(\boldsymbol{\theta}|y_1, y_2, \ldots, y_n\ ) \\
&\quad = \quad P(Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n|\boldsymbol{\theta}) \\
&\qquad\quad \text{(by definition of likelihood function)} \\
&\quad = \quad P(Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n, \boldsymbol{T} = \boldsymbol{t}|\boldsymbol{\theta}) \\
&\qquad\quad \text{(as the two events are same)} \\
&\quad = \quad P(\boldsymbol{T} = \boldsymbol{t}|\boldsymbol{\theta})P(Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n|\boldsymbol{T} = \boldsymbol{t}, \boldsymbol{\theta}) \\
&\qquad\quad \text{(by definition of conditional probability.)} \\
&\quad = \quad g(\boldsymbol{t}, \boldsymbol{\theta})h(y_1, y_2, \ldots, y_n\ )
\end{aligned}$$

where $g(\boldsymbol{t}, \theta) = P(\boldsymbol{T} = \boldsymbol{t}|\boldsymbol{\theta})$, which involves only $\boldsymbol{\theta}$ and $\boldsymbol{t}$ (without directly involving $y_1, y_2, \ldots, y_n$ - the only way $y_1, y_2, \ldots, y_n$ appear in the expression of $g(\boldsymbol{t}, \boldsymbol{\theta})$ is through $\boldsymbol{t}$); and $h(y_1, y_2, \ldots, y_n\ ) = P(Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n|\boldsymbol{T} = \boldsymbol{t}, \boldsymbol{\theta})$, which does not involve $\boldsymbol{\theta}$, by definition of the sufficiency of $\boldsymbol{T}$.
<u>"if" or $\Longleftarrow$ part:</u> Suppose the probability model of $Y_1, Y_2, \ldots, Y_n$ is such that it admits the factorisation (6). Then

$$\begin{aligned}
&P(Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n|\boldsymbol{T} = \boldsymbol{t}, \boldsymbol{\theta}) \\
&\quad = \quad \frac{P(Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n, \boldsymbol{T} = \boldsymbol{t}|\boldsymbol{\theta})}{P(\boldsymbol{T} = \boldsymbol{t}|\boldsymbol{\theta})} \\
&\quad = \quad \frac{g(\boldsymbol{t}, \theta)h(y_1, y_2, \ldots, y_n\ )}{\sum_{\{y_1, y_2, \ldots, y_n\ :\boldsymbol{T}(y_1, y_2, \ldots, y_n\ )=\boldsymbol{t}\}} g(\boldsymbol{t}, \boldsymbol{\theta})h(y_1, y_2, \ldots, y_n\ )}
\end{aligned}$$

$$= \frac{g(\boldsymbol{t},\boldsymbol{\theta})h(y_1,y_2,\ldots,y_n\ )}{g(\boldsymbol{t},\boldsymbol{\theta})\sum_{\{y_1,y_2,\ldots,y_n\ :\boldsymbol{T}(y_1,y_2,\ldots,y_n\ )=\boldsymbol{t}\}} h(y_1,y_2,\ldots,y_n\ )}$$

$$= \frac{h(y_1,y_2,\ldots,y_n\ )}{\sum_{y_1,y_2,\ldots,y_n\ :\boldsymbol{T}(y_1,y_2,\ldots,y_n\ )=\boldsymbol{t}} h(y_1,y_2,\ldots,y_n\ )}$$

which does not depend on $\boldsymbol{\theta}$. $\bigtriangledown$

<u>Example 5:</u> A. i. Consider the Normal likelihood given in (3). Assume that σ^2 is known but not μ. Let $T(Y_1,Y_2,\ldots,Y_n\ )=\overline{Y}$, so that $t=\overline{y}$. Then the likelihood function (3) can be factorised into $g(t,\mu)=e^{-(t-\mu)^2/(2\sigma^2)}$ and $h(y_1,y_2,\ldots,y_n\ )=(2\pi\sigma^2)^{-n/2}e^{-\frac{1}{2\sigma^2}\sum_{i=1}^n(y_i-\overline{y})^2}$, showing that $\overline{Y}$ is sufficient for μ in a $N(\mu,\sigma^2)$ model for known σ^2.

A. ii. Again consider the Normal likelihood given in (3). This time assume that μ is known but not σ^2. Let $T(Y_1,Y_2,\ldots,Y_n\ )=\sum_{i=1}^n(Y_i-\mu)^2$. Note that this T is a statistic because μ is known. In this case define $g(t,\sigma^2)=(2\pi\sigma^2)^{-n/2}e^{-\frac{1}{2\sigma^2}t}$ and $h(y_1,y_2,\ldots,y_n\ )=1$ so that $L(\sigma^2|y_1,y_2,\ldots,y_n\ )=g(t,\sigma^2)h(y_1,y_2,\ldots,y_n\ )$. Thus in this case $\sum_{i=1}^n(Y_i-\mu)^2$ is sufficient for σ^2.

A. iii. Finally consider the Normal likelihood in (3) with both (μ,σ^2) unknown. Note that in this case the unknown parameter is vector valued with $\boldsymbol{\theta}=(\mu,\sigma^2)$. In this case we should have a vector valued sufficient statistics $\boldsymbol{T}$. Thus let $\boldsymbol{T}=(\overline{Y},\sum_{i=1}^n(Y_i-\overline{Y})^2)$, $g(\boldsymbol{t},\boldsymbol{\theta})=(2\pi\sigma^2)^{-n/2}e^{-\frac{1}{2\sigma^2}\left(\sum_{i=1}^n(y_i-\overline{y})^2+(\overline{y}-\mu)^2\right)}$ and $h(y_1,y_2,\ldots,y_n\ )=1$, so that $L(\boldsymbol{\theta}|y_1,y_2,\ldots,y_n\ )=g(\boldsymbol{t},\boldsymbol{\theta})h(y_1,y_2,\ldots,y_n\ )$. Thus in this case $(\overline{Y},\sum_{i=1}^n(Y_i-\overline{Y})^2)$ is sufficient for $\boldsymbol{\theta}=(\mu,\sigma^2)$.

B. For the Bernoulli likelihood in (4) let $T=\sum_{i=1}^n Y_i$, $g(t,p)=p^t(1-p)^{n-t}$ and $h(y_1,y_2,\ldots,y_n\ )=1$. Then $L(p|y_1,y_2,\ldots,y_n\ )=g(t,p)h(y_1,y_2,\ldots,y_n\ )$ and thus $\sum_{i=1}^n Y_i$ is sufficient for p.

C. $\sum_{i=1}^n Y_i$ is sufficient for λ of the Poisson model, because for the Poisson model define $T=\sum_{i=1}^n Y_i$, $g(t,\lambda)=e^{-n\lambda}\lambda^t$, and $h(y_1,y_2,\ldots,y_n\ )=1/\prod_{i=1}^n y_i!$ so that the Poisson likelihood in (5) equals $g(t,\lambda)h(y_1,y_2,\ldots,y_n\ )$. $\bigtriangledown$

We are now in a position to state the first theorem pertaining UMVUE, due to Rao and Blackwell, which is as follows.

<u>Theorem 3</u> (Rao-Blackwell Theorem): If T is sufficient for θ and U is any unbiased estimate of θ, then the statistic $h(T)=E[U|T]$ is also unbiased for θ and $V_\theta[h(T)]\leq V_\theta[U]$.

<u>Proof:</u> First note that $h(T)$, the conditional expectation of U given T is a statistic *i.e.* can be completely determined from the data, because since T is sufficient, the conditional distribution of $Y_1,Y_2,\ldots,Y_n\ |T$ and hence $E[U(Y_1,Y_2,\ldots,Y_n\ )|T]=h(T)$ also does not depend on θ.

Next observe that $h(T)$ is unbiased for θ. This is because $E_\theta[h(T)]=E_\theta[E[U|T]]=E_\theta[U]=\theta\ \forall\theta\in\Theta$. The last equality follows because U is unbiased for θ, and the one before that follows because $E[E[X|Y]]=E[X]$. Now,

$$\begin{aligned}
&V_\theta[U]\\
&\quad= E_\theta[(U-\theta)^2]\\
&\qquad\text{(because } U \text{is unbiased for } \theta)
\end{aligned}$$

$$
\begin{aligned}
&= E_\theta[\{(U-h(T))+(h(T)-\theta)\}^2] \\
&= E_\theta[(U-h(T))^2]+E_\theta[(h(T)-\theta)^2]+2E_\theta[(U-h(T))(h(T)-\theta)] \\
&= E_\theta[(U-h(T))^2]+V_\theta[h(T)]+2E_\theta[E[\{(U-h(T))(h(T)-\theta)|T]] \\
&\quad \text{(because } h(T)\text{is unbiased for } \theta \text{ and } E[E[X|Y]=E[X]) \\
&= E_\theta[(U-h(T))^2]+V_\theta[h(T)]+2E_\theta[(h(T)-\theta)E[(U-h(T))|T]] \\
&\quad \text{(because given } T,(h(T)-\theta)\text{is a constant w.r.t. } T \text{ and hence can be taken out of the} \\
&\quad \text{expectation)} \\
&= E_\theta[(U-h(T))^2]+V_\theta[h(T)] \\
&\quad \text{(because } E[(U-h(T))|T]]=E[U|T]-E[h(T)|T]=h(T)-h(T)=0) \\
&\geq V_\theta[h(T)] \\
&\quad \text{(because } E_\theta[(U-h(T))^2]\geq 0) \qquad \bigtriangledown
\end{aligned}
$$

Rao-Blackwell theorem only goes on to show how an arbitrary unbiased estimator may be improved w.r.t. reducing its variance by preserving its unbiasedness. It states that the way to reduce variance of an unbiased estimator after preserving its unbiasedness would be to consider a new statistic which is the conditional expectation of the unbiased estimator given a sufficient statistics. Incidentally this process of taking conditional expectation of a statistic given a sufficient statistics is called Rao-Blackwellisation. As such the theorem does not directly state anything about an estimator being a UMVUE. Indeed according to the theorem, if U_1 and U_2 are two unbiased estimators of θ then they can respectively be improved upon by considering $h_1(T)=E[U_1|T]$ and $h_2(T)=E[U_2|T]$, but it does not say anything comparing the variances of $h_1(T)$ and $h_2(T)$. This problem is resolved by introducing another concept called **completeness**.

<u>Definition 5:</u> A statistic T is called **complete** if $E_\theta[g(T)]=0\ \forall\theta\in\Theta \Longrightarrow g(T)\equiv 0$.[3]

[3]The intuition behind completeness of a statistic can be understood only together with sufficiency. Opposite of a sufficient statistic is called an **ancillary** statistic. A statistic is called **ancillary** if its distribution does not depend on the unknown θ. For example if $Y_1,Y_2,\ldots,Y_n$ are i.i.d. Uniform$[\theta-1/2,\theta+1/2]$ or Uniform$[\theta,\theta+1]$, the smallest and largest observations denoted by $(Y_{(1)},Y_{(n)})$ are sufficient (use factorisation theorem) but $Y_{(n)}-Y_{(1)}$ is ancillary in either case. A sufficient statistic packs in itself all the information the data has got about the unknown θ, while an ancillary statistics is useless in drawing any inference about θ since its distribution does not depend on θ. Now a sufficient statistic is most successful in optimal data reduction when there is no ancillary statistics. In all the instances of Example 5, the number of parameters and the corresponding number of sufficient statistics agree, because in those cases there do not exist any ancillary statistic. But in the Uniform$[\theta-1/2,\theta+1/2]$ or Uniform$[\theta,\theta+1]$ examples above, the sufficient statistics is two-dimensional for a single parameter θ. This is because of the presence of the ancillary statistic. Now a weaker notion of ancillarity, called first-order ancillarity, occurs when the expectation of a non-constant statistic does not depend on θ. Ancillarity requires the entire distribution be free from θ while first-order ancillarity requires the same only about its mean. Completeness of a sufficient statistic guarantees that there is no first-order ancillary statistics and thus no ancillary statistics (because ancillarity implies first-order ancillarity). Because if T is complete and some function of it has a constant expectation c no matter what the value of θ is, then the function must identically equal c (if $E_\theta[f(T)]=c\ \forall\theta\in\Theta$, $E_\theta[f(T)-c]=0\ \forall\theta\in\Theta$ implying $f(T)\equiv c$ by completeness). Thus if a sufficient statistic is also complete it does not contain any ancillarity and as a result we can expect good results from functions of this sufficient statistic such as an UMVUE for example.

<u>Theorem 4</u> (Lehmann-Scheffe Theorem): If T is a complete sufficient statistic then for any unbiased estimate U of θ, $h(T) = E[U|T]$ is the unique UMVUE for θ.

<u>Proof:</u> We have already shown that if T is sufficient, for any arbitrary unbiased estimate U of θ, $V_\theta[h(T)] \leq V_\theta[U]$, where $h(T) = E[U|T]$. Now if U_1 and and U_2 are two different unbiased estimators of θ and $h_1(T) = E[U_1|T]$ and $h_2(T) = E[U_2|T]$, then since $E_\theta[h_1(T)] = E_\theta[h_2(T)] = \theta \ \forall\theta \in \Theta$, $E_\theta[h_1(T) - h_2(T)] = 0 \ \forall\theta \in \Theta$, which implies $h_1(T) \equiv h_2(T)$, since T is complete. Thus no matter whatever unbiased estimator U one may start with, after its Rao-Blackwellisation w.r.t. a complete sufficient statistic T, one always ends up with the same or unique Rao-Blackwellised version $h(T)$ whose variance is no greater than any arbitrary unbiased estimator of θ. Thus $h(T)$ is the unique UMVUE for θ. $\bigtriangledown$

<u>Example 6:</u> A random variable Y is said to belong to the **exponential family of distributions** if it has the p.d.f. (or p.m.f)

$$f(y|\boldsymbol{\eta}) = \exp\{\sum_{i=1}^{k} \eta_i T_i(y) + A(\boldsymbol{\eta}) + B(y)\} \tag{7}$$

where $\boldsymbol{\eta} = (\eta_1, \eta_2, \ldots, \eta_k)$ is the vector of unknown parameters and $A(\cdot)$, $B(\cdot)$, $T_1(\cdot)$, $T_2(\cdot)$, $\ldots$, $T_k(\cdot)$, are known functions.

P.d.f/p.m.f of many standard probability models can be expressed in the form given in (7).

If $Y \sim B(n, p)$, the Binomial distribution with p as the parameter of interest and known n,

$$p(y|p) = \begin{pmatrix} n \\ y \end{pmatrix} p^y (1-p)^{n-y} = \exp\left\{\eta y + n \log\left(\frac{1}{1+e^\eta}\right) + \log\begin{pmatrix} n \\ y \end{pmatrix}\right\}$$

where $\eta = \log\left(\frac{p}{1-p}\right)$. This is in the form of (7) with $k = 1$, $T_1(y) = y$, $A(\eta) = n \log\left(\frac{1}{1+e^\eta}\right)$ and $B(y) = \log\begin{pmatrix} n \\ y \end{pmatrix}$.

If $Y \sim N(\mu, \sigma^2)$, the Normal distribution with (μ, σ^2) as the unknown parameters,

$$\begin{aligned} & f(y|\mu, \sigma^2) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y-\mu)^2\right\} \\ &= \exp\left\{\left(\eta_1 y + \eta_2 y^2\right) + \frac{1}{2}\left(\frac{\eta_1^2}{2\eta_2} + \log(-\eta_2) - \log\pi\right)\right\} \end{aligned}$$

where $\eta_1 = \frac{\mu}{\sigma^2}$ and $\eta_2 = -\frac{1}{2\sigma^2}$. This is in the form of (7) with $k = 2$, $T_1(y) = y$, $T_2(y) = y^2$, $A(\boldsymbol{\eta}) = \frac{1}{2}\left(\frac{\eta_1^2}{2\eta_2} + \log(-\eta_2) - \log\pi\right)$ and $B(y) = 0$.

If $Y \sim \text{Poisson}(\lambda)$, $p(y|\lambda) = e^{-\lambda}\frac{\lambda^y}{y!} = \exp\{\eta y - e^\eta - \log y!\}$, where $\eta = \log\lambda$. This is in the form of (7) with $k = 1$, $T_1(y) = y$, $A(\eta) = -e^\eta$ and $B(y) = \log y!$.

If $Y \sim \text{Gamma}(\alpha, \lambda)$, with both α and λ as unknown parameters,

$$f(y|\alpha, \lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\lambda y} = \exp\{(\eta_1 \log y + \eta_2 y) + (\eta_1 \log(-\eta_2) - \log\Gamma(\eta_1)) - \log y\}$$

where $\eta_1 = \alpha$ and $\eta_2 = -\lambda$. This is in the form of (7) with $k = 2$, $T_1(y) = \log y$, $T_2(y) = y$, $A(\boldsymbol{\eta}) = \eta_1 \log(-\eta_2) - \log \Gamma(\eta_1)$ and $B(y) = -\log y$.

Similarly, Negative Binomial, Hypergeometric and Beta distributions can also be shown to belong to the exponential family.[4] Note that the parameter $\boldsymbol{\eta}$ in (7) is not the natural parameter of interest for the standard models as discussed above. It is called the canonical parameter. However in all cases it will be the case that there exists a one-to-one relationship between the natural parameter of a model and the canonical parameter, as illustrated in the above important special models. We need not worry too much about this issue. As shall be seen shortly, we shall be able to draw inference about the natural parameter of interest only. Introduction of the concept of exponential family only facilitates the theoretical development in the sense that it allows us to prove results like sufficiency and completeness and hence the UMVUE for many probability models of interest at once in a unified manner.

Sufficiency: If the distribution of Y belongs to the exponential family, and $Y_1, Y_2, \ldots, Y_n$ is an i.i.d. sample of Y, then it is immediate that $\boldsymbol{T}(Y_1, Y_2, \ldots, Y_n) = (\sum_{i=1}^n T_1(Y_i), \ldots, \sum_{i=1}^n T_k(Y_i))$ is sufficient for $\boldsymbol{\eta}$ using the Factorisation Theorem. This is because, in this case the likelihood of $\boldsymbol{\eta}$

$$\begin{aligned} & L(\boldsymbol{\eta}|y_1, y_2, \ldots, y_n) \qquad (8) \\ &= \exp\left\{\sum_{j=1}^k \eta_j \sum_{i=1}^n T_j(y_i) + nA(\boldsymbol{\eta})\right\} \exp\left\{\sum_{i=1}^n B(y_i)\right\} \\ &= g(\boldsymbol{t}, \boldsymbol{\eta}) h(y_1, y_2, \ldots, y_n) \end{aligned}$$

where $g(\boldsymbol{t}, \boldsymbol{\eta}) = \exp\left\{\sum_{j=1}^k \eta_j \sum_{i=1}^n T_j(y_i) + nA(\boldsymbol{\eta})\right\}$ and $h(y_1, y_2, \ldots, y_n) = \exp\{\sum_{i=1}^n B(y_i)\}$.

Completeness: We shall prove completeness of the exponential family only for $k = 1$, which is easily generalisable to higher dimension. Suppose the distribution of Y belongs to the exponential family, and $E_{\boldsymbol{\eta}}[g(Y)] = 0 \ \forall \boldsymbol{\eta}$. Then from (7) it follows that $\int \{g(y) \exp(B(y))\} \exp(\eta y) dy$[5] $= 0 \ \forall \eta$. But the l.h.s. is nothing but the Laplace transformation of $g(y) \exp(B(y))$. Since Laplace transformation of a function is unique, $g(y) \exp(B(y)) \equiv 0$, implying $g(y) \equiv 0$.[6]

Thus for exponential family by the Rao-Blackwell and Lehmann-Scheffe theorems any function of $\boldsymbol{T}$ is the UMVUE of its expectation. This is because the conditional expectation of any function of $\boldsymbol{T}$ given $\boldsymbol{T}$ is that function itself, and any statistic is an unbiased estimate of its own expectation. Now since $\boldsymbol{T}$ is also complete, it is the UMVUE.

[4] However not all important probability models, useful in application belongs to the exponential family. For example, the Weibull distribution with p.d.f. $f(y|\lambda, \beta) = \lambda\beta y^{\beta-1} e^{-\lambda y^\beta}$ can at best be simplified to $\exp\left\{\left(\beta \log y - \lambda e^{\beta \log y}\right) + (\log \lambda + \log \beta) - \log y\right\}$, which is not in the form given in (7).

[5] In case Y is discrete this integral is to be replaced by summation and the rest of the argument carries through.

[6] Completeness of the exponential family should be understood in terms the distribution of the corresponding sufficient statistics $\boldsymbol{T}$. Say for example if $Y \sim N(0, \sigma^2)$ then it belongs to the exponential family. But Y is not complete. This is because for example $E_{\sigma^2}[Y^\alpha] = 0 \ \forall \sigma^2 > 0 \ \forall \alpha = 1, 3, 5, \ldots$ odd powers. Or for that matter expectation of any function of Y, which is symmetric about 0 is 0. But those functions are not 0, showing that Y is not complete. But in this case the sufficient statistics is Y^2, which of course is complete.

Thus if $Y \sim B(n,p)$, since $E_p[Y/n] = p$ and Y is complete and sufficient, the sample proportion $\hat{p} = Y/n$ is the UMVUE of the population proportion p.

If $Y_1, Y_2, \ldots, Y_n$ are i.i.d. $N(\mu, \sigma^2)$, $(\sum_{i=1}^n Y_i, \sum_{i=1}^n Y_i^2)$ is complete and sufficient for $(\mu/\sigma^2, -1/(2\sigma^2))$. But since $(\sum_{i=1}^n Y_i, \sum_{i=1}^n Y_i^2) \leftrightarrow (\overline{Y} = \frac{1}{n}\sum_{i=1}^n Y_i, s_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^n (Y_i - \overline{Y})^2)$ and $(\mu/\sigma^2, -1/(2\sigma^2)) \leftrightarrow (\mu, \sigma^2)$ is one-to-one the sample mean and variance $(\overline{Y}, s_{n-1}^2)$ is complete and sufficient for (μ, σ^2). Now since $E_{\mu,\sigma^2}[\overline{Y}] = \frac{1}{n}\sum_{i=1}^n E_{\mu,\sigma^2}[Y_i] = \frac{1}{n}\sum_{i=1}^n \mu = \mu$, $\overline{Y}$ is the UMVUE of μ. Similarly since $E_{\mu,\sigma^2}[s_{n-1}^2] = \frac{1}{n-1}E_{\mu,\sigma^2}\left[\sum_{i=1}^n (Y_i - \overline{Y})^2\right] = \frac{1}{n-1}E_{\mu,\sigma^2}\left[\sum_{i=1}^n Y_i^2 - n\overline{Y}^2\right] = \left[\frac{1}{n-1}\sum_{i=1}^n \left\{V_{\mu,\sigma^2}[Y_i] + (E_{\mu,\sigma^2}[Y_i])^2\right\}\right] - \frac{n}{n-1}\left\{V_{\mu,\sigma^2}[\overline{Y}] + \left(E_{\mu,\sigma^2}[\overline{Y}]\right)^2\right\}$ (because for any random variable X, $E[X^2] = V[X] + (E[X])^2) = \left[\frac{1}{n-1}\sum_{i=1}^n (\sigma^2 + \mu^2)\right] - \frac{n}{n-1}\left(\frac{\sigma^2}{n} + \mu^2\right)$ (because $V_{\mu,\sigma^2}[\overline{Y}] = \frac{1}{n^2}\sum_{i=1}^n V_{\mu,\sigma^2}[Y_i] = \frac{1}{n^2}\sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}$, and we have just shown that $E_{\mu,\sigma^2}[\overline{Y}] = \mu$.) $= \left(\frac{n}{n-1} - \frac{1}{n-1}\right)\sigma^2 = \sigma^2$, s_{n-1}^2 is the UMVUE of σ^2.

If $Y_1, Y_2, \ldots, Y_n$ are i.i.d. Poisson(λ) then $\sum_{i=1}^n Y_i$ is complete and sufficient. Since $E_\lambda\left[\sum_{i=1}^n Y_i\right] = n\lambda$ $\overline{Y} = \frac{1}{n}\sum_{i=1}^n Y_i$ is the UMVUE of λ. $\bigtriangledown$

We move on to the other method of obtaining UMVUE after presenting an interesting example of obtaining UMVUE through Rao-Blackwellisation.

Example 7: If T denotes the life of a system, its reliability at time t is defined as $P[T > t]$. Exponential distribution is a very popular model for life of a system. So let T have an exponential distribution with mean θ, denoted by $T \sim \exp(\theta)$ and suppose we are interested in estimating the reliability of the system at a give mission time t. For this purpose suppose we observe time to failure or life of n independent systems, $T_1 = t_1, \ldots, T_n = t_n$. Now if $T \sim \exp(\theta)$, $P[T > t] = e^{-t/\theta}$, and $\overline{T} = \frac{1}{n}\sum_{i=1}^n T_i$ is the UMVUE of θ, because $\sum_{i=1}^n T_i$ is a complete and sufficient statistic for $\exp(\theta)$ and $E_\theta\left[\frac{1}{n}\sum_{i=1}^n T_i\right] = \theta$. But this does not imply that $e^{-t/\overline{T}}$ is the UMVUE for $e^{-t/\theta}$. In fact $e^{-t/\overline{T}}$ is not even an unbiased estimate of $e^{-t/\theta}$ and this leads to the search for an UMVUE of $e^{-t/\theta}$.

With the first observation T_1, the estimator $I_t(T_1) = \begin{cases} 1 & \text{if } T_1 > t \\ 0 & \text{otherwise} \end{cases}$, the indicator function of the event $T_1 > t$, is an unbiased estimate of $P[T > t]$ for all life distribution T (not just exponential), because $E_\theta[I_t(T_1)] = P[T > t]$. Now for exponential distribution since $\sum_{i=1}^n T_i$ is a complete and sufficient statistic for θ, by Rao-Blackwell Lehmann-Scheffe theorems $E\left[I_t(T_1) | \sum_{i=1}^n T_i\right]$ would be the UMVUE for $P[T > t]$. The derivation of this quantity only requires some probability calculation which is as follows.

Let $S = \sum_{i=1}^n T_i$ and $R = \sum_{i=2}^n T_i$. Since $I_t(T_1)$ is a function of T_1, its conditional expectation is found by deriving the conditional density $f_{T_1|S}(t_1|s)$ of $T_1|S$, which requires the joint density $f_{T_1,S}(t_1, s)$ of (T_1, S). This is accomplished by considering the joint density of (T_1, R) (which is easy because of the independence of T_1 and R) and then considering the one-to-one onto transformation $(T_1, R) \leftrightarrow (T_1, S)$ defined as $T_1 = T_1$ and $S = R + T_1$. Note that $S \sim \text{Gamma}(n, 1/\theta)$ and likewise $R \sim \text{Gamma}(n-1, 1/\theta)$. Thus the density of R is

$$f_R(r) = \frac{1}{\theta^{n-1}(n-2)!} r^{n-2} e^{-r/\theta}$$

and since $T_1 \sim \exp(\theta)$ the joint density of (T_1, R) is given by

$$f_{T_1,r}(t_1, r) = \frac{1}{\theta^n (n-2)!} r^{n-2} e^{-(r+t_1)/\theta}.$$

Now applying the one-to-one onto transformation $(T_1, R) \leftrightarrow (T_1, S)$ we get the joint density of (T_1, S) as

$$f_{T_1,S}(t_1, s) = \frac{1}{\theta^n (n-2)!} (s - t_1)^{n-2} e^{-s/\theta}.$$

The conditional density $f_{T_1|S}(t_1|s)$ of $T_1|S$ is found by dividing $f_{T_1,S}(t_1, s)$ by the marginal density $f_S(s)$ of S, which since is a Gamma$(n, 1/\theta)$ density,

$$f_{T_1|S}(t_1|s) = \frac{n-1}{s} \left(1 - \frac{t_1}{s}\right)^{n-2} \qquad 0 < t_1 < s.$$

Now since $I_t(T_1)$ is the indicator function of the event $T_1 > t$, $E\left[I_t(T_1)|S\right]$ is found by integrating $f_{T_1|S}(t_1|s)$ from t to t_1's upper-bound s. This integral is found as follows.

$$\begin{aligned}
&E\left[I_t(T_1)|S\right] \\
&= \frac{n-1}{s} \int_t^s \left(1 - \frac{t_1}{s}\right)^{n-2} \; dt_1 \\
&= (n-1) \int_0^{1-t/s} u^{n-2} \; du \\
&\quad \text{(with the substitution } u = 1 - t_1/s \text{ we have } du = -dt_1/s \text{ and the range of the integral} \\
&\quad \text{from } 1 - t/s \text{ to 0, which after adjusting for the negative sign becomes from 0 to } 1 - t/s.) \\
&= \left(1 - \frac{t}{s}\right)^{n-1}
\end{aligned}$$

Thus the UMVUE of $P[T > t] = e^{-t/\theta}$, the reliability at mission time t of a system having an exponential life distribution with mean θ, with observations $T_1, \ldots, T_n$ is given by $\left(1 - \frac{t}{\sum_{i=1}^n T_i}\right)^{n-1}$. $\bigtriangledown$

Now we shall present the second and final method of determining an UMVUE. Though so far while discussing the theory, we have pretended as if our interest lies only in the natural population parameter $\boldsymbol{\theta}$, as Example 7 reveals, many times we might be interested in some function $\phi(\boldsymbol{\theta})$ of the original population parameter θ. Note that this does not require redoing the previous Rao-Blackwell Lehmann-Scheffe theory of determining the UMVUE. In this case one confines oneself with unbiased estimators of $\phi(\boldsymbol{\theta})$ and restricts one's search of UMVUE only among functions of complete sufficient statistics of $\boldsymbol{\theta}$.

So now suppose we are interested in estimating some parametric function $\phi(\theta)$ of θ. In this discussion we shall only deal with scalar valued θ and $\phi(\theta)$.

<u>Theorem 5</u> (Cramer-Rao Theorem): If the population p.d.f. $f(y|\theta)$ is "regular", $Y_1, Y_2, \ldots, Y_n$ are i.i.d. $f(y|\theta)$ and $T(Y_1, Y_2, \ldots, Y_n)$ is an unbiased estimate of $\phi(\theta)$, then $V_\theta[T] \geq -\frac{\{\phi'(\theta)\}^2}{\left\{nE_\theta\left[\frac{\partial^2 \log f(Y|\theta)}{\partial \theta^2}\right]\right\}}$.[7]

[7] In the discrete case, replace the p.d.f. $f(y|\theta)$ by p.m.f. $p(y|\theta)$ in the statement of the theorem and replace the integrals by summations in its proof, and the same result follows through.

<u>Proof:</u> Let $\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_n)$ and $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)$. Since the joint density of $Y_1, Y_2, \ldots, Y_n$ is same as the likelihood function of θ, as defined in Definition 4, $E_\theta\left[g(\boldsymbol{Y})\right] = \int_{\Re^n} g(\boldsymbol{y}) L(\theta|\boldsymbol{y})\, d\boldsymbol{y}$ for any function of $\boldsymbol{Y}$ and

$$\int_{\Re^n} L(\theta|\boldsymbol{y})\, d\boldsymbol{y} = 1.$$

Partially differentiating the above w.r.t. θ within the integral sign (which is allowed since the model is assumed to be "regular") we get

$$\int_{\Re^n} \frac{\partial L(\theta|\boldsymbol{y})}{\partial \theta}\, d\boldsymbol{y} = 0$$

which can be rewritten as

$$E_\theta\left[\frac{\partial \log L(\theta|\boldsymbol{Y})}{\partial \theta}\right] = \int_{\Re^n} \left\{\frac{1}{L(\theta|\boldsymbol{y})}\frac{\partial L(\theta|\boldsymbol{y})}{\partial \theta}\right\} L(\theta|\boldsymbol{y})\, d\boldsymbol{y} = 0 \tag{9}$$

If we again differentiate (9) w.r.t. θ under the integral sign (which is allowed under a second "regularity" condition) we get

$$\begin{aligned}
&\int_{\Re^n} \left\{\left(\frac{1}{L(\theta|\boldsymbol{y})}\frac{\partial L(\theta|\boldsymbol{y})}{\partial \theta}\right)\frac{\partial L(\theta|\boldsymbol{y})}{\partial \theta} + L(\theta|\boldsymbol{y})\frac{\partial}{\partial \theta}\left(\frac{1}{L(\theta|\boldsymbol{y})}\frac{\partial L(\theta|\boldsymbol{y})}{\partial \theta}\right)\right\} d\boldsymbol{y} \\
&= \int_{\Re^n} \left\{\left(\frac{\partial \log L(\theta|\boldsymbol{y})}{\partial \theta}\right)\left(\frac{1}{L(\theta|\boldsymbol{y})}\frac{\partial L(\theta|\boldsymbol{y})}{\partial \theta}\right) + \frac{\partial}{\partial \theta}\left(\frac{\partial \log L(\theta|\boldsymbol{y})}{\partial \theta}\right)\right\} L(\theta|\boldsymbol{y})\, d\boldsymbol{y} \\
&= \int_{\Re^n} \left\{\left(\frac{\partial \log L(\theta|\boldsymbol{y})}{\partial \theta}\right)^2 + \frac{\partial^2 \log L(\theta|\boldsymbol{y})}{\partial \theta^2}\right\} L(\theta|\boldsymbol{y})\, d\boldsymbol{y} = 0
\end{aligned}$$

The last equality implies that

$$E_\theta\left[\left(\frac{\partial \log L(\theta|\boldsymbol{Y})}{\partial \theta}\right)^2\right] = -E_\theta\left[\frac{\partial^2 \log L(\theta|\boldsymbol{Y})}{\partial \theta^2}\right]. \tag{10}$$

Now if $T(\boldsymbol{Y})$ is an unbiased estimator of $\phi(\theta)$

$$\int_{\Re^n} T(\boldsymbol{y}) L(\theta|\boldsymbol{y})\, d\boldsymbol{y} = \phi(\theta). \tag{11}$$

Differentiating (11) under the integral sign (which is allowed under a third "regularity" condition) we obtain

$$\int_{\Re^n} T(\boldsymbol{y})\frac{\partial \log L(\theta|\boldsymbol{y})}{\partial \theta} L(\theta|\boldsymbol{y})\, d\boldsymbol{y} = \phi'(\theta)$$

and by (9)

$$\int_{\Re^n} \left\{T(\boldsymbol{y}) - \phi(\theta)\right\}\frac{\partial \log L(\theta|\boldsymbol{y})}{\partial \theta} L(\theta|\boldsymbol{y})\, d\boldsymbol{y} = \phi'(\theta). \tag{12}$$

Hence by (9), (11) and (12)

$$Cov_\theta\left[T(\boldsymbol{Y}), \frac{\partial \log L(\theta|\boldsymbol{Y})}{\partial \theta}\right] = \phi'(\theta). \tag{13}$$

Now by (9) and (10)

$$\begin{aligned} & V_\theta\left[\frac{\partial \log L(\theta|\boldsymbol{Y})}{\partial\theta}\right] \\ = & -E_\theta\left[\frac{\partial^2 \log L(\theta|\boldsymbol{Y})}{\partial\theta^2}\right] = -E_\theta\left[\frac{\partial^2}{\partial\theta^2}\left\{\sum_{i=1}^n \log(f(Y_i|\theta))\right\}\right] \\ = & -nE_\theta\left[\frac{\partial^2 \log(f(Y|\theta))}{\partial\theta^2}\right]. \end{aligned} \tag{14}$$

Let $\rho_{T(\mathbf{Y}),\frac{\partial \log L(\theta|\mathbf{Y})}{\partial\theta}}$ denote the correlation coefficient between $T(\boldsymbol{Y})$ and $\frac{\partial \log L(\theta|\mathbf{Y})}{\partial\theta}$. Then by (13) and (14)

$$\rho^2_{T(\mathbf{Y}),\frac{\partial \log L(\theta|\mathbf{Y})}{\partial\theta}} = \frac{\left\{Cov_\theta\left[T(\boldsymbol{Y}),\frac{\partial \log L(\theta|\mathbf{Y})}{\partial\theta}\right]\right\}^2}{\left(V_\theta\left[T(\boldsymbol{Y})\right]\right)\left(V_\theta\left[\frac{\partial \log L(\theta|\mathbf{Y})}{\partial\theta}\right]\right)} = \frac{\{\phi'(\theta)\}^2}{\left(V_\theta\left[T(\boldsymbol{Y})\right]\right)\left(-nE_\theta\left[\frac{\partial^2 \log(f(Y|\theta)}{\partial\theta^2}\right]\right)} \leq 1$$

implying $V_\theta[T] \geq -\frac{\{\phi'(\theta)\}^2}{\left\{nE_\theta\left[\frac{\partial^2 \log f(Y|\theta)}{\partial\theta^2}\right]\right\}}$. $\bigtriangledown$

The quantity $-\frac{\{\phi'(\theta)\}^2}{\left\{nE_\theta\left[\frac{\partial^2 \log f(Y|\theta)}{\partial\theta^2}\right]\right\}}$ is called the **Cramer-Rao lower bound (CRLB)**. The quantity $-E_\theta\left[\frac{\partial^2 \log f(Y|\theta)}{\partial\theta^2}\right]$ which by virtue of (10) (for $n = 1$) equals $E_\theta\left[\left(\frac{\partial \log f(Y|\theta)}{\partial\theta}\right)^2\right]$ is called **Fisher Information** and is denoted by $\mathcal{I}(\theta)$. Fisher Information for the entire sample, denoted by $\mathcal{I}_n(\theta)$, which in general is given in (10) reduces to $n\mathcal{I}(\theta)$ in the i.i.d. case. The intuitive reason behind calling this quantity "information" will be clear in the next sub-subsection when we take up the Method of Maximum Likelihood or Maximum Likelihood Estimation (MLE). In this connection note that the CRLB will be attained by an unbiased estimator $T(\boldsymbol{Y})$ of $\phi(\theta)$ if and only if its correlation with $\frac{\partial \log L(\theta|\boldsymbol{Y})}{\partial\theta}$ is ± 1 which happens if and only if $T(\boldsymbol{Y}) - \phi(\theta)$ is a constant multiple of $\frac{\partial \log L(\theta|\boldsymbol{Y})}{\partial\theta}$, since by (9) $E_\theta\left[\frac{\partial \log L(\theta|\boldsymbol{Y})}{\partial\theta}\right] = 0$. But in general since this constant multiple may depend on θ it may be said that CRLB is attained by an unbiased estimator $T(\boldsymbol{Y})$ of $\phi(\theta)$ if and only if

$$\frac{\partial \log L(\theta|\boldsymbol{Y})}{\partial\theta} = A(\theta)\left\{T(\boldsymbol{Y}) - \phi(\theta)\right\} \tag{15}$$

We shall close this sub-subsection by illustrating the utility of CRLB in determination of UMVUE. Theorem 5 gives a lower-bound for the variance of an unbiased estimator $T(\boldsymbol{Y})$ of a parametric function $\phi(\theta)$. Thus if we can somehow catch hold of an unbiased estimator $\hat{\phi}$ of $\phi(\theta)$, whose variance coincides with this lower-bound, then we know that the variance of any other unbiased estimator is more than that of $\hat{\phi}$ and thus $\hat{\phi}$ is a UMVUE. That is if the CRLB is attained by the variance of any unbiased estimator of $\phi(\theta)$, then it is a UMVUE.

<u>**Example 8:**</u> **A.** If $Y_1, Y_2, \ldots, Y_n$ are i.i.d. Bernoulli(p)

$$\mathcal{I}(p) = V_p\left[\frac{\partial}{\partial p}\left\{Y\log(p) + (1-Y)\log(1-p)\right\}\right] = V_p\left[\frac{Y}{p(1-p)}\right] = \frac{1}{p(1-p)}$$

and $V_p\left[\hat{p} = \frac{1}{n}\sum_{i=1}^n Y_i\right] = \frac{p(1-p)}{n} = \frac{1}{\mathcal{I}_n(p)}$, the CRLB, showing the sample proportion to be the UMVUE for the population proportion.

B. Let $Y_1, Y_2, \ldots, Y_n$ be i.i.d. $N(\mu, 1)$. Then

$$\mathcal{I}_n(\mu) = -E_\mu\left[\frac{\partial^2}{\partial\mu^2}\log\left(\prod_{i=1}^n\left\{\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(Y_i-\mu)^2}\right\}\right)\right] = -E_\mu\left[\frac{\partial}{\partial\mu}\sum_{i=1}^n(Y_i-\mu)\right] = n$$

and $V_\mu[\overline{Y}] = 1/n = 1/\mathcal{I}_n(\mu)$. Since $\overline{Y}$ is unbiased for μ it is an UMVUE, as its variance is attaining the CRLB.

C. Let $Y_1, Y_2, \ldots, Y_n$ be i.i.d. exponential with mean θ. Then $f(y|\theta) = (1/\theta)e^{-y/\theta}$. Thus $\frac{\partial^2}{\partial\theta^2}\log f(y|\theta) = \frac{\partial}{\partial\theta}\{-(1/\theta)+(y/\theta^2)\} = (1/\theta^2)-2(y/\theta^3)$, so that $\mathcal{I}(\theta) = -E_\theta\left[(1/\theta^2) - 2(Y/\theta^3)\right] = (1/\theta^2)$ and $\mathcal{I}_n(\theta) = n/\theta$. Now if we are interested in estimating θ since $E_\theta[\overline{Y}] = \theta$ and $V_\theta[\overline{Y}] = \theta^2/n = 1/\mathcal{I}_n(\theta) =$ the CRLB, it is the UMVUE for θ. $\bigtriangledown$

2.1.2 MLE

So far we have discussed some criteria under which an estimator may be called "good" and have only provided at best some indirect methods of obtaining such estimators. In general it is desirable to have some automatic methods which yield estimators which are at least approximately "good". One such method is the method of Maximum Likelihood (ML) yielding an MLE. If a UMVUE does not exist or cannot be obtained using the results discussed in §2.1.1, then by far the most popular method of estimation employed in practice is the method of Maximum Likelihood. This is partly because instead of having an assortment of theorems guiding one in the search of a "good" estimator as in the case of UMVUE, it is an automatic method producing at least numerical estimates; and partly (actually philosophically the only reason) because the method yields estimators which are "good" in the asymptotic sense and thus work very well for large samples.

Let the parameter of interest $\boldsymbol{\theta}$ be vector valued, and let $L(\boldsymbol{\theta}|\boldsymbol{y})$ denote the likelihood of $\boldsymbol{\theta}$ given the observations $\boldsymbol{y}$, where the likelihood function $L(\boldsymbol{\theta}|\boldsymbol{y})$ is as has been defined in Definition 4.

<u>**Definition 6:**</u> $\hat{\boldsymbol{\theta}}$ is called the **Maximum Likelihood Estimator** (**MLE**) of $\boldsymbol{\theta}$ if $L(\hat{\boldsymbol{\theta}}|\boldsymbol{y}) \geq L(\boldsymbol{\theta}|\boldsymbol{y})\ \forall\boldsymbol{\theta} \in \boldsymbol{\Theta}$.

Intuitively, granting the informal interpretation of the likelihood function discussed in the paragraph following Definition 4, MLE $\hat{\boldsymbol{\theta}}$ is that value of $\boldsymbol{\theta}$ for which one is most likely to observe a data set such as the one at hand namely $\boldsymbol{Y} = \boldsymbol{y}$, among all possible values of $\boldsymbol{\theta} \in \boldsymbol{\Theta}$. Other than this intuitive appeal, in the so-called "regular"[8] cases, computation of MLE is conceptually straight forward. All one has to do is find the maxima of $L(\boldsymbol{\theta}|\boldsymbol{y})$, which is typically accomplished by solving the system of equations $\left.\frac{\partial L(\boldsymbol{\theta}|\boldsymbol{y})}{\partial\boldsymbol{\theta}}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} = \mathbf{0}$ or equivalently by solving

$$\left.\frac{\partial\ell(\boldsymbol{\theta}|\boldsymbol{y})}{\partial\boldsymbol{\theta}}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} = \mathbf{0} \text{ where } \ell(\boldsymbol{\theta}|\boldsymbol{y}) = \log(L(\boldsymbol{\theta}|\boldsymbol{y})) \tag{16}$$

[8] A model $f(y|\boldsymbol{\theta})$ or $p(y|\boldsymbol{\theta})$ is regular if it satisfies the conditions required for interchanging the differentiation and integration signs in Theorem 5 yielding CRLB.

((16) is usually easier to deal with because the likelihood function is a product which after taking log becomes a sum), and then showing the resulting solution $\hat{\boldsymbol{\theta}}$ to be the global maxima[9].

<u>**Example 9:**</u> **A.** If $Y_1, Y_2, \ldots, Y_n$ are i.i.d. Bernoulli(p)

$$L(p|\boldsymbol{y}) = p^{\sum_{i=1}^n y_i}(1-p)^{n-\sum_{i=1}^n y_i} \Rightarrow \ell(p|\boldsymbol{y}) = \log\left(\frac{p}{1-p}\right)\sum_{i=1}^n y_i + n\log(1-p)$$

and

$$\left.\frac{\partial \ell(p|\boldsymbol{y})}{\partial p}\right|_{p=\hat{p}} = 0 \Rightarrow \hat{p} = \frac{1}{n}\sum_{i=1}^n y_i$$

Since $\frac{\partial^2 \ell(p|\boldsymbol{y})}{\partial p^2} = -\left(\frac{\sum_{i=1}^n y_i}{p^2} + \frac{n-\sum_{i=1}^n y_i}{(1-p)^2}\right) < 0 \ \forall 0 < p < 1$, $\ell(p|\boldsymbol{y})$ is concave and thus $\hat{p} = \frac{1}{n}\sum_{i=1}^n y_i$, the sample proportion is the MLE of the unknown population proportion p.

B. Let $Y_1, Y_2, \ldots, Y_n$ be i.i.d. $N(\mu, \sigma^2)$. Then

$$L(\mu,\sigma^2) = (2\pi\sigma^2)^{-n/2}\exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^n (y_i-\mu)^2\right\} \Rightarrow \ell(\mu,\sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^n (y_i-\mu)^2$$

Now

$$\begin{pmatrix} \left.\frac{\partial \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial \mu}\right|_{(\mu=\hat{\mu},\sigma^2=\widehat{\sigma^2})} \\ \left.\frac{\partial \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial (\sigma^2)}\right|_{(\mu=\hat{\mu},\sigma^2=\widehat{\sigma^2})} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} \sum_{i=1}^n (y_i-\hat{\mu}) \\ \frac{\sum_{i=1}^n (y_i-\hat{\mu})^2}{2(\widehat{\sigma^2})^2} \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{n}{2\widehat{\sigma^2}} \end{pmatrix} \Rightarrow \begin{pmatrix} \hat{\mu} \\ \widehat{\sigma^2} \end{pmatrix} = \begin{pmatrix} \overline{y} \\ \frac{1}{n}\sum_{i=1}^n (y_i-\overline{y})^2 \end{pmatrix}.$$

The second derivative matrix of $\ell(\mu,\sigma^2|\boldsymbol{y})$

$$\begin{bmatrix} \frac{\partial^2 \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial \mu^2} & \frac{\partial^2 \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial \mu \partial (\sigma^2)} \\ \frac{\partial^2 \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial (\sigma^2)\partial \mu} & \frac{\partial^2 \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial (\sigma^2)^2} \end{bmatrix} = -\begin{bmatrix} \frac{n}{\sigma^2} & \frac{1}{(\sigma^2)^2}\sum_{i=1}^n (y_i-\mu) \\ \frac{1}{(\sigma^2)^2}\sum_{i=1}^n (y_i-\mu) & \frac{1}{(\sigma^2)^2}\left\{n/2 - \frac{1}{\sigma^2}\sum_{i=1}^n (y_i-\mu)^2\right\} \end{bmatrix}$$

is negative-definite at $(\hat{\mu}, \widehat{\sigma^2})$ and is asymptotically so as $n \to \infty$ in probability[10]. Because by the law of large numbers

$$\frac{1}{n}\begin{bmatrix} \frac{\partial^2 \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial \mu^2} & \frac{\partial^2 \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial \mu \partial (\sigma^2)} \\ \frac{\partial^2 \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial (\sigma^2)\partial \mu} & \frac{\partial^2 \ell(\mu,\sigma^2|\boldsymbol{y})}{\partial (\sigma^2)^2} \end{bmatrix} \xrightarrow{P} -\begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & -\frac{1}{(\sigma^2)^2} \end{bmatrix}.$$

Hence $(\hat{\mu}, \widehat{\sigma^2}) = (\overline{y}, \frac{1}{n}\sum_{i=1}^n (y_i-\overline{y})^2)$ is the MLE of (μ, σ^2). $\triangledown$

Having understood the method *i.e.* how to implement it, let us now turn our attention to the more important question of the justification of this method. That is why in a given

[9] Showing something to be a global maxima in general is a daunting task, but in most MLE computations this feat is typically accomplished by showing the log-likelihood is concave.

[10] A sequence of random variables $\{X_n\}_{n=1}^\infty$ is said to converge to a random variable X **in probability**, notationally represented as $X_n \xrightarrow{P} X$, if $\forall \epsilon > 0$ the sequence of real numbers $P[|X_n - X| < \epsilon] \to 1$ as $n \to \infty$.

situation MLE should be considered and what optimality property does it possess? We shall present the proofs only in the single parameter case, which are easily extendable to multi or vector-valued parameter case. Also we shall only be concerned with regular cases. For non-regular cases ML may or may not be the best approach for estimation. But before discussing optimality of MLE, we have to first introduce another criterion for an estimator being "good" called **consistency**.

Definition 7: An estimator $\hat{\theta}_n$, based on a sample of size n is said to be **consistent** for θ if $\theta_n \xrightarrow{P} \theta \ \forall \theta \in \Theta$.

That is if an estimator is consistent, it has the desirable property of converging to the true unknown value as the sample size increases.

Example 10 (Weak Law of Large Numbers): Let $\overline{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$ denote the sample mean based on n i.i.d. observations $Y_1, Y_2, \ldots, Y_n$ from an arbitrary population having mean μ and variance σ^2. Since $E[\overline{Y}_n] = \mu$ (Example 2) and $V[\overline{Y}_n] = \sigma^2/n$ (Example 6, page 13), by Chebyshev's inequality

$$P\left[|\overline{Y}_n - \mu| < \epsilon\right] = P\left[|\overline{Y}_n - \mu| < \sqrt{n}\frac{\epsilon}{\sigma}\left(\frac{\sigma}{\sqrt{n}}\right)\right] \geq 1 - \frac{\sigma^2}{n\epsilon^2} \to 1 \text{ as } n \to \infty$$

Thus the sample mean is always a consistent estimate of population mean, provided the population mean and variance exists. $\bigtriangledown$

Theorem 6: Assume the probability model is "regular". Then as $n \to \infty$, (16) always has a solution with probability tending to 1. Furthermore such a solution is unique with probability tending to 1, which is also consistent for $\boldsymbol{\theta}$.

Proof: Above theorem essentially has three claims namely existence of a solution, uniqueness of the solution and finally consistency of the solution. We shall prove existence, consistency and uniqueness, in that order, after some preliminaries.

Let $\boldsymbol{\theta}_0$ denote the true unknown value of the parameter of interest $\boldsymbol{\theta}$. Consider the random variable $\frac{L(\boldsymbol{\theta}|\boldsymbol{Y})}{L(\boldsymbol{\theta}_0|\boldsymbol{Y})}$, where $L(\boldsymbol{\theta}|\boldsymbol{Y})$ is the Likelihood function of the data set $\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_n)$. Since arithmetic mean is always greater than geometric mean, so are their logarithms, which means logarithm of an expectation or arithmetic mean is always greater than the expectation of the logarithms (because that's what the logarithm of a geometric mean reduces to). Thus,

$$\log\left(E_{\boldsymbol{\theta}_0}\left[\frac{L(\boldsymbol{\theta}|\boldsymbol{Y})}{L(\boldsymbol{\theta}_0|\boldsymbol{Y})}\right]\right) > E_{\boldsymbol{\theta}_0}\left[\log\left(\frac{L(\boldsymbol{\theta}|\boldsymbol{Y})}{L(\boldsymbol{\theta}_0|\boldsymbol{Y})}\right)\right]$$

where $E_{\boldsymbol{\theta}_0}[g(\boldsymbol{Y})]$ denotes the expectation of the random variable $g(\boldsymbol{Y})$ when $\boldsymbol{\theta}$ equals its true unknown value $\boldsymbol{\theta}_0$. Now $\forall \boldsymbol{\theta} \in \boldsymbol{\Theta}$,

$$E_{\boldsymbol{\theta}_0}\left[\frac{L(\boldsymbol{\theta}|\boldsymbol{Y})}{L(\boldsymbol{\theta}_0|\boldsymbol{Y})}\right] = \int_{\Re^n} \frac{L(\boldsymbol{\theta}|\boldsymbol{y})}{L(\boldsymbol{\theta}_0|\boldsymbol{y})}\prod_{i=1}^{n} f(y_i|\boldsymbol{\theta}_0)d\boldsymbol{y} = \int_{\Re^n} \frac{L(\boldsymbol{\theta}|\boldsymbol{y})}{L(\boldsymbol{\theta}_0|\boldsymbol{y})}L(\boldsymbol{\theta}_0|\boldsymbol{y})d\boldsymbol{y} = \int_{\Re^n}\prod_{i=1}^{n} f(y_i|\boldsymbol{\theta})d\boldsymbol{y} = 1$$

Thus $\forall \boldsymbol{\theta} \in \boldsymbol{\Theta}$,

$$E_{\boldsymbol{\theta}_0}\left[\log\left(L(\boldsymbol{\theta}_0|\boldsymbol{Y})\right)\right] \geq E_{\boldsymbol{\theta}_0}\left[\log\left(L(\boldsymbol{\theta}|\boldsymbol{Y})\right)\right]$$

But $\frac{1}{n}\log\left(L(\boldsymbol{\theta}|\boldsymbol{Y})\right) = \frac{1}{n}\sum_{i=1}^{n}\log\left(f(Y_i|\boldsymbol{\theta})\right)\ \forall\boldsymbol{\theta}\in\boldsymbol{\Theta}$, and thus by the law of large numbers (see Example 10 above) with probability tending to 1, $\frac{1}{n}\log\left(L(\boldsymbol{\theta}|\boldsymbol{Y})\right)$ would equal $E_{\boldsymbol{\theta}_0}\left[\log\left(L(\boldsymbol{\theta}|\boldsymbol{Y})\right)\right]$. Hence $\forall\boldsymbol{\theta}\in\boldsymbol{\Theta}$,

$$\log\left(L(\boldsymbol{\theta}_0|\boldsymbol{Y})\right) \geq \log\left(L(\boldsymbol{\theta}|\boldsymbol{Y})\right) \text{ with probability tending to 1.} \tag{17}$$

Existence: All the statements below are true with probability tending to 1. According to (17), $\log\left(L(\boldsymbol{\theta}|\boldsymbol{Y})\right)$ or $\ell(\boldsymbol{\theta}|\boldsymbol{Y})$ will have a maxima at $\boldsymbol{\theta}_0$. Hence its derivative must vanish at $\boldsymbol{\theta}_0$. Therefore equation (16) must have a solution (at $\boldsymbol{\theta}_0$ in particular). This shows that there exists a solution for equation (16).

Consistency: Substituting $\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}$, the MLE of $\boldsymbol{\theta}$, in (17) we get $\log\left(L(\boldsymbol{\theta}_0|\boldsymbol{Y})\right) \geq \log\left(L(\hat{\boldsymbol{\theta}}|\boldsymbol{Y})\right)$ with probability tending to 1. But on the other hand, by the definition of MLE (*vide.* Definition 6), $\log\left(L(\hat{\boldsymbol{\theta}}|\boldsymbol{Y})\right) \geq \log\left(L(\boldsymbol{\theta}|\boldsymbol{Y})\right)\ \forall\boldsymbol{\theta}\in\boldsymbol{\Theta}$, and thus in particular when $\boldsymbol{\theta}=\boldsymbol{\theta}_0$, $\log\left(L(\hat{\boldsymbol{\theta}}|\boldsymbol{Y})\right) \geq \log\left(L(\boldsymbol{\theta}_0|\boldsymbol{Y})\right)$. Therefore the MLE $\hat{\boldsymbol{\theta}}$ have to coincide with $\boldsymbol{\theta}_0$, the true unknown value of the parameter of interest, with probability tending to 1, as $n\to\infty$, showing that MLE is consistent.

Uniqueness: We shall outline the proof for the single parameter case. The proof for the general multi-parameter case is analogous, but since it requires a little more additional notations is not included here. MLE is obtained by solving equation (16) and then checking whether it is at least a local maxima or not (because equation (16) would be satisfied by both local minima as well as saddle points of the log likelihood function $\ell(\theta|\boldsymbol{Y})$) by looking at the sign of $\left.\frac{\partial^2\ell(\theta|\boldsymbol{Y})}{\partial\theta^2}\right|_{\theta=\hat{\theta}}$, which needs to be negative for $\hat{\theta}$ to be a maxima. Let $\hat{\theta}_c$ be any consistent estimator of θ. Then by the law of large numbers

$$\frac{1}{n}\left.\frac{\partial^2\ell(\theta|\boldsymbol{Y})}{\partial\theta^2}\right|_{\theta=\hat{\theta}_c} = \frac{1}{n}\sum_{i=1}^{n}\left.\frac{\partial^2\log(f(Y_i|\theta))}{\partial\theta^2}\right|_{\theta=\hat{\theta}_c} \xrightarrow{P} E_{\theta_0}\left[\left.\frac{\partial^2\log(f(Y|\theta))}{\partial\theta^2}\right|_{\theta=\theta_0}\right]$$

Now by (10) (for $n=1$) $\forall\theta\in\Theta$,

$$E_\theta\left[\frac{\partial^2\log(f(Y|\theta))}{\partial\theta^2}\right] = -E_\theta\left[\left(\frac{\partial\log(f(Y|\theta))}{\partial\theta}\right)^2\right] < 0$$

and hence $E_{\theta_0}\left[\left.\frac{\partial^2\log(f(Y|\theta))}{\partial\theta^2}\right|_{\theta=\theta_0}\right] < 0$, which means that if $\hat{\theta}_c$ is a consistent estimator then by the law of large numbers, with probability tending to 1, $\left.\frac{\partial^2\ell(\theta|\boldsymbol{Y})}{\partial\theta^2}\right|_{\theta=\hat{\theta}_c} < 0$. Now let $\hat{\theta}_1<\hat{\theta}_2$ be two distinct MLE's. Then they also must be consistent. Furthermore since the regularity conditions guarantee that likelihood function is smooth there must exist a minima $\hat{\theta}_3$ such that $\hat{\theta}_1<\hat{\theta}_3<\hat{\theta}_2$. Since $\hat{\theta}_3$ is a minima $\left.\frac{\partial^2\ell(\theta|\boldsymbol{Y})}{\partial\theta^2}\right|_{\theta=\hat{\theta}_3} > 0$. But since $\hat{\theta}_1$ and $\hat{\theta}_2$ are consistent, and $\hat{\theta}_1<\hat{\theta}_3<\hat{\theta}_2$ so will be $\hat{\theta}_3$, in which case $\left.\frac{\partial^2\ell(\theta|\boldsymbol{Y})}{\partial\theta^2}\right|_{\theta=\hat{\theta}_3} < 0$, which is a contradiction, and thus one cannot have more than one MLE. $\bigtriangledown$

In the foregoing proof of uniqueness, we saw that $\left.\frac{\partial^2 \ell(\theta|\boldsymbol{Y})}{\partial \theta^2}\right|_{\theta=\hat{\theta}}$ played a crucial role in establishing an estimate to be an MLE. It was also seen that this quantity approximately equals $-E_\theta\left[\frac{\partial^2 \ell(\theta|\boldsymbol{Y})}{\partial \theta^2}\right] = \mathcal{I}_n(\theta)$ with $\theta = \theta_0$. The quantity $\mathcal{I}(\theta) = \frac{\infty}{\backslash}\mathcal{I}_\backslash(\theta)$ also arose in §2.2.1 in the context of CRLB, where $\mathcal{I}(\theta)$ was called Fisher Information. It is called "information" because of the following reasons stated below (without proof):

(i) By (10)

$$\mathcal{I}(\theta) = -E_\theta\left[\frac{\partial^2 \ell(\theta|Y)}{\partial \theta^2}\right] = E_\theta\left[\left(\frac{\partial \ell(\theta|Y)}{\partial \theta}\right)^2\right] > 0 \;\; \forall \theta \in \Theta$$

(ii) $\mathcal{I}(\theta)$ is additive in the sense that if $\mathcal{I}^{(1)}(\theta)$ and $\mathcal{I}^{(2)}(\theta)$ respectively denote informations for two samples Y_1 and Y_2 of size 1 each, then the information $\mathcal{I}_2(\theta)$ for the combined sample $\{Y_1, Y_2\}$ of size 2 equals $\mathcal{I}^{(1)}(\theta) + \mathcal{I}^{(2)}(\theta)$, provided of course they are independent.

(iii) If $\mathcal{I}^{(T)}(\theta)$ denotes information based on a statistic $T(Y)$, then $\mathcal{I}(\theta) \geq \mathcal{I}^{(T)}(\theta)$, with equality iff T is sufficient. This shows that you always loose some information by summarising raw data, unless it is summarised using sufficient statistics, in which case you do not loose any information as from the original raw data.

(iv) A variety of distance measures between two distributions for two different values of θ has an approximate increasing relationship with $\mathcal{I}(\theta)$. What this means is $\mathcal{I}(\theta)$ can be viewed as a measure of sensitivity of changing the population distribution by changing θ.

Apart from the above mathematical facts, the so called **likelihood principle** says that whatever a data set $\boldsymbol{Y}$ has to say about the unknown parameter θ, it is packed in the likelihood or log-likelihood function $\ell(\theta|\boldsymbol{Y})$. Now for the same parameter for two different data sets the one with sharper (more peaked) $\ell(\theta|\boldsymbol{Y})$ is more informative about θ. That is flatter the likelihood lesser is the information about θ. $\mathcal{I}_n(\theta)(= n\mathcal{I}(\theta))$, negative of the expected value of the second derivative of the log-likelihood simply measures how peaked one can expect this log-likelihood function to be. The negative sign simply ensures that the quantity is positive. Larger this value the likelihood is expected to be more peaked and thus more is the information about θ and hence the name "information".

Fisher Information being an expectation is a population quantity, and thus depicts the kind of informative behaviour to expect from a likelihood, in general for an arbitrary sample from this population. But for a given set of data the peakedness of the likelihood can be exactly measured without resorting to its expected value. This consideration gives rise to the sample version of Fisher information called the **Observed Information** which is given by $I_n(\theta) = -\frac{\partial^2 \ell(\theta|\boldsymbol{Y})}{\partial \theta^2}$. Note that by law of large numbers $\forall \theta \in \Theta$, $I(\theta) = \frac{1}{n} I_n(\theta) \xrightarrow{P} \mathcal{I}(\theta)$, which allows us to consistently estimate the Fisher Information by Observed Information.

In the multi-parameter or vector valued $\boldsymbol{\theta}$, instead of a scalar Fisher/ observed information we have a **Fisher Information Matrix** $\boldsymbol{\mathcal{I}}(\boldsymbol{\theta})$ and its sample analogue **Observed**

Information Matrix $\boldsymbol{I}(\boldsymbol{\theta})$ which are as follows. Let $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_k)$ have k components. Then

$$\boldsymbol{\mathcal{I}}(\boldsymbol{\theta}) = -\begin{bmatrix} E_{\boldsymbol{\theta}}\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|Y)}{\partial \theta_1^2}\right] & E_{\boldsymbol{\theta}}\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|Y)}{\partial \theta_1 \partial \theta_2}\right] & \cdots & E_{\boldsymbol{\theta}}\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|Y)}{\partial \theta_1 \partial \theta_k}\right] \\ E_{\boldsymbol{\theta}}\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|Y)}{\partial \theta_2 \partial \theta_1}\right] & E_{\boldsymbol{\theta}}\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|Y)}{\partial \theta_2^2}\right] & \cdots & E_{\boldsymbol{\theta}}\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|Y)}{\partial \theta_2 \partial \theta_k}\right] \\ \vdots & \vdots & \ddots & \vdots \\ E_{\boldsymbol{\theta}}\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|Y)}{\partial \theta_k \partial \theta_1}\right] & E_{\boldsymbol{\theta}}\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|Y)}{\partial \theta_k \partial \theta_2}\right] & \cdots & E_{\boldsymbol{\theta}}\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|Y)}{\partial \theta_k^2}\right] \end{bmatrix}$$

and likewise

$$\boldsymbol{I}(\boldsymbol{\theta}) = -\begin{bmatrix} \frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell(\boldsymbol{\theta}|Y_i)}{\partial \theta_1^2} & \frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell(\boldsymbol{\theta}|Y_i)}{\partial \theta_1 \partial \theta_2} & \cdots & \frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell(\boldsymbol{\theta}|Y_i)}{\partial \theta_1 \partial \theta_k} \\ \frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell(\boldsymbol{\theta}|Y_i)}{\partial \theta_2 \partial \theta_1} & \frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell(\boldsymbol{\theta}|Y_i)}{\partial \theta_2^2} & \cdots & \frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell(\boldsymbol{\theta}|Y_i)}{\partial \theta_2 \partial \theta_k} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell(\boldsymbol{\theta}|Y_i)}{\partial \theta_k \partial \theta_1} & \frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell(\boldsymbol{\theta}|Y_i)}{\partial \theta_k \partial \theta_2} & \cdots & \frac{1}{n}\sum_{i=1}^n \frac{\partial^2 \ell(\boldsymbol{\theta}|Y_i)}{\partial \theta_k^2} \end{bmatrix}.$$

As far as the optimality of MLE is concerned, so far we have only shown that it is consistent. Though often MLE is the starting point of the search of an UMVUE its relationship with an UMVUE still remains to be examined. Actually the connection between the two in the single parameter case is best studied through Theorem 5, establishing the CRLB. Thus suppose there exists an unbiased estimator $T(\boldsymbol{Y})$ for a parametric function $\phi(\theta)$, which attains the CRLB. Then by (15) and (16) $\hat{\theta}$, the MLE of θ must satisfy $T(\boldsymbol{Y}) = \phi(\hat{\theta})$ which goes on to show that the UMVUE of $\phi(\theta)$ is a function of the MLE. In case $\phi(\theta) \equiv \theta$ and $T(\boldsymbol{Y})$ is unbiased for θ attaining CRLB, the above argument also shows that then $T(\boldsymbol{Y})$ must equal the MLE $\hat{\theta}$.

The connection between UMVUE and MLE in the multi-parameter case is easiest to appreciate for the exponential family of distributions, covering most of the probability models applied in practice, introduced in Example 6. Again as usual we shall show the mathematics for the continuous case, which carries through exactly the same way in the discrete case by replacing the p.d.f. by p.m.f. and the integrals by summations. By (7) since $f(y|\boldsymbol{\eta})$ is a p.d.f.

$$\int_{\Re} \exp\left\{\sum_{l=1}^k \eta_l T_l(y) + B(y)\right\} dy = e^{-A(\boldsymbol{\eta})}.$$

Differentiating the above w.r.t. η_j, interchanging the differentiation and integral sign, denoting $\frac{\partial A(\boldsymbol{\eta})}{\partial \eta_j}$ by $A_j(\boldsymbol{\eta})$, and then multiplying both sides by $e^{-A(\boldsymbol{\eta})}$, we get$\forall j = 1, 2, \ldots, k$,

$$\int_{\Re} T_j(y) \exp\left\{\sum_{l=1}^k \eta_l T_l(y) + B(y) + A(\boldsymbol{\eta})\right\} dy = -A_j(\boldsymbol{\eta})$$

or $\int_{\Re} T_j(y) f(y|\boldsymbol{\eta})\ dy = E_{\boldsymbol{\eta}}[T_j(Y)] = -A_j(\boldsymbol{\eta})$. Thus $T(Y)$ is an unbiased estimator of $-A_j(\boldsymbol{\eta})$, and as shown in Example 6, based on a sample of size n, $\frac{1}{n}\sum_{i=1}^n T_j(Y_i)$ is the UMVUE for $-A_j(\boldsymbol{\eta})$. But by (9) the likelihood equation (16) in this case reduces to

$$\frac{1}{n}\sum_{i=1}^n T_j(Y_i) = -A_j(\hat{\boldsymbol{\eta}}) \ \ \forall j = 1, 2, \ldots, k,$$

where $\hat{\boldsymbol{\eta}}$ is the MLE of $\boldsymbol{\eta}$ showing that functions of $\boldsymbol{T}$ being UMVUE estimates of their expectations (*vide.* Example 6) are necessarily functions of the MLE $\hat{\boldsymbol{\eta}}$.

We shall close our discussion about MLE after proving an important theorem. This theorem, among other things, like enabling one to approximate the sampling distribution of the MLE in large samples, also helps one to see the connection between MLE and CRLB.

<u>Theorem 7:</u> For "regular" models, for large n, the sampling distribution of MLE $\hat{\boldsymbol{\theta}}$ may be approximated by a k-variate Normal distribution with mean $\boldsymbol{\theta}_0$ and variance-covariance matrix $\boldsymbol{\mathcal{I}}_n^{-1}(\boldsymbol{\theta}_0)$, where $\boldsymbol{\theta}_0$ is the true unknown value of $\boldsymbol{\theta}$.

<u>Proof:</u> We shall give the outline of the proof only for $k = 1$. For general k, though the proof is similar, is a little more technical in nature and hence omitted. By Taylor's theorem, expanding the l.h.s. of the likelihood equation (16) about θ_0, the true unknown value of θ, we get,

$$\left.\frac{\partial \ell(\theta|\boldsymbol{Y})}{\partial \theta}\right|_{\theta=\hat{\theta}} = \left.\frac{\partial \ell(\theta|\boldsymbol{Y})}{\partial \theta}\right|_{\theta=\theta_0} + (\hat{\theta} - \theta_0) \left.\frac{\partial^2 \ell(\theta|\boldsymbol{Y})}{\partial \theta^2}\right|_{\theta=\theta^*}$$

where θ^* lies in between $\hat{\theta}$ and θ_0. But by (16) since the l.h.s. of the above equation is 0,

$$(\hat{\theta} - \theta_0) = \left.\frac{\partial \ell(\theta|\boldsymbol{Y})}{\partial \theta}\right|_{\theta=\theta_0} \Big/ \left\{ - \left.\frac{\partial^2 \ell(\theta|\boldsymbol{Y})}{\partial \theta^2}\right|_{\theta=\theta^*} \right\}.$$

Multiplying both sides of above by $\sqrt{n\mathcal{I}(\theta_0)}$, and multiplying the numerator and denominator of the r.h.s. by $\frac{1}{n}$ this equation may be rewritten as

$$(\hat{\theta} - \theta_0)\sqrt{n\mathcal{I}(\theta_0)} = \frac{\left\{ \frac{1}{n} \left.\frac{\partial \ell(\theta|\boldsymbol{Y})}{\partial \theta}\right|_{\theta=\theta_0} \right\} \Big/ \sqrt{\mathcal{I}(\theta_0)/n}}{\left\{ \frac{1}{n} \left.\frac{\partial^2 \ell(\theta|\boldsymbol{Y})}{\partial \theta^2}\right|_{\theta=\theta^*} \right\} \Big/ \{-\mathcal{I}(\theta_0)\}}. \tag{18}$$

Now by law of large numbers, definition of $\mathcal{I}(\theta_0)$, and because θ^* is sandwitched between θ_0 and $\hat{\theta}$, with $\hat{\theta} \xrightarrow{P} \theta_0$,

$$\frac{1}{n} \left.\frac{\partial^2 \ell(\theta|\boldsymbol{Y})}{\partial \theta^2}\right|_{\theta=\theta^*} = \frac{1}{n} \sum_{i=1}^{n} \left.\frac{\partial^2 \log f(Y_i|\theta)}{\partial \theta^2}\right|_{\theta=\theta^*} \xrightarrow{P} -\mathcal{I}(\theta_0).$$

Thus the denominator of (18) converges in probability to 1. For handling the numerator note that

$$\frac{1}{n} \left.\frac{\partial \ell(\theta|\boldsymbol{Y})}{\partial \theta}\right|_{\theta=\theta_0} = \frac{1}{n} \sum_{i=1}^{n} U_i$$

where $U_i = \left.\frac{\partial \log f(Y_i|\theta)}{\partial \theta}\right|_{\theta=\theta_0}$. By (9), $E_{\theta_0}[U_i] = 0$ and by (9) and (10), $V_{\theta_0}[U_i] = \mathcal{I}(\theta_0) = \sigma^2$ (say). Thus by denoting $\frac{1}{n}\sum_{i=1}^{n} U_i$ by $\overline{U}$, the numerator of (18) may be written as $\frac{\overline{U}-0}{\sqrt{\sigma^2/n}}$ whose distribution would approach that of a standard Normal distribution as $n \to \infty$ by the Central Limit Theorem. Thus the distribution of $(\hat{\theta} - \theta_0)\sqrt{n\mathcal{I}(\theta_0)}$, the l.h.s. of (18),

must also approach that of a standard Normal distribution as $n \to \infty$, implying that $\hat{\theta}$ is asymptotically Normal with mean θ_0 and variance $1/\{n\mathcal{I}(\theta_0)\}$. $\bigtriangledown$

Above theorem shows that the asymptotic variance of MLE is same as the CRLB. In this sense MLE may be called an "efficient" estimator. However a general discussion of **efficiency** is beyond the scope of this notes, and thus we shall wrap up our discussion about MLE by making a a couple of final remarks.

Note that optimality properties of MLE like consistency and asymptotic Normality with CRLB as the asymptotic variance are all large sample properties. That is they are true only if the sample size $n \to \infty$. Thus it must be borne in mind that one can expect MLE to behave reasonably well only when the sample size is large, unless other results show them to be optimal for small samples.

The second point is regarding numerical computation. Since more often than not the likelihood equation (16) does not admit closed form analytical solution, one has to employ numerical methods. In such situations one typically employs standard numerical method like Newton-Raphson to directly solve (16), or utilise method of steepest ascent for direct maximisation of $\ell(\boldsymbol{\theta}|\boldsymbol{Y})$. Newton-Raphson requires the second derivative matrix of $\ell(\boldsymbol{\theta}|\boldsymbol{Y})$. When one uses the expression of the expected values of these second derivatives in the the Newton-Raphson algorithm, the method is called Rao's method of scoring, which typically converges faster than the usual Newton-Raphson.

$1/\sqrt{n\mathcal{I}(\theta_0)}$ is called the asymptotic standard error of MLE. Since one does not know θ_0 the true unknown value of θ one may replace θ_0 by MLE $\hat{\theta}$ in the expression for $\mathcal{I}(\theta)$ or alternatively report $1/\sqrt{I_n(\hat{\theta})}$ as estimated asymptotic standard error of the MLE, which also is a consistent estimator. As shall be shortly seen in §2.2, the section on interval estimation, standard error is typically best interpreted in its intuitive sense when the sampling distribution of the estimator is Normal. In case of MLE, by virtue of Theorem 7 since it is so, this asymptotic standard error typically is a fairly interpretable quantity whose use and interpretation will be further clarified in §2.2.

2.1.3 Other Methods

For a given estimation problem though we first strive to procure a UMVUE with the search possibly guided by first obtaining an anlytical expression for MLE, because of the computational difficulties sometimes one resorts to some other alternative methods. In this subsection we shall discuss a few such other methods.

Method of Moments (MM)

In this method given an i.i.d. sample $Y_1, Y_2, \ldots, Y_n$ from the population one equates the sample moment(s) with the theoretical population moment(s) which are typically some functions of the original population parameters of interest $\boldsymbol{\theta}$. If $\boldsymbol{\theta}$ has k components one typically considers the first k raw or central moments, whichever is convenient. Let μ_r denote the r-th raw moment in the population and $\overline{Y}_r$ denote the corresponding r-th raw moment in

the sample. Then

$$\mu_r = \int_{\Re} y^r f(y|\boldsymbol{\theta})\, dy = g_r(\boldsymbol{\theta}) \text{ (say) and } \overline{Y}_r = \frac{1}{n}\sum_{i=1}^{n} Y_i^r$$

Now MM requires one to form the system of k equations

$$g_r(\boldsymbol{\theta}) = \overline{Y}_r \text{ for } r = 1, 2, \ldots, k$$

which are then solved for $\boldsymbol{\theta}$ to obtain MM estimator $\hat{\boldsymbol{\theta}}_M$ of $\boldsymbol{\theta}$.

Example 11: Let $Y_1, Y_2, \ldots, Y_n$ be i.i.d. Gamma(α, λ) with population p.d.f. $\frac{\lambda^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\lambda y}$. Now as can be immediately seen, method of ML would require messing around with the di-gamma function (derivative of $\log \Gamma()$ functions), which is not exactly a routine numerical task. But by appealing to the method of moments we immediately observe that the population mean or the first raw moment equals α/λ and the population variance or the second central moment equals α/λ^2. Equating these two to their respective (unbiased) sample counter-parts $\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ and $s_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2$ and solving for (α, λ), we immediately obtain the MM estimates of (α, λ) as $\hat{\alpha}_M = \overline{Y}^2/s_{n-1}^2$ and $\hat{\lambda}_M = \overline{Y}/s_{n-1}^2$. $\bigtriangledown$

Now let us address the issue of optimality of MM. By law of large numbers $\overline{Y}_r \xrightarrow{P} \mu_r$. Now if the $g_r(\cdot)$ functions are such that $(g_1(\boldsymbol{\theta}), \ldots, g_k(\boldsymbol{\theta})) \longleftrightarrow (\theta_1, \ldots, \theta_k)$ is one-to-one with continuous inverse (as in the gamma example above), then $\hat{\boldsymbol{\theta}}_M$ is a consistent estimator of $\boldsymbol{\theta}$. Furthermore $\overline{Y}_r$'s are unbiased estimators of μ_r's. (But this does not imply that $\hat{\boldsymbol{\theta}}_M$ is unbiased for $\boldsymbol{\theta}$, unless of course if the $g_r(\cdot)$'s are linear, which is extremely rare.) Other than these there is no other compleing reason to use MM. In fact MM estimators in general are less efficient than MLE, where the notion of **efficiency** was very briefly touched upon in §2.1.2.